## Experimental Benchmarking Data Tables, Charts, and Graphs

### 1) Encryption Stats

The encryption stats dataset captures the high-precision execution latency which is measured in milliseconds of individual cryptographic phases during the Enochian encryption process. Telemetry was collected using synchronous garbage collection and hardware-level **Stopwatch** APIs to isolate the computational overhead of distinct transformations, including subset-sum key encapsulation, plaintext sanitization, modular shifts, and radix-21 alphabet mapping. The data presents the latency from 2 x 2 to 10 x 10 dimensional matrix across exponentially scaling payload volumes to demonstrate the algorithmic scaling bounds.

### 2 x 2 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 20000 Words (Maximum) |
|---|---|---|---|---|---|
| Receiver Public Key Generation: | 3.8985 | 356.9693 | 16.6323 | 129.9177 | 13.6606 |
| Step 1: Session Setup | 13.6851 | 18.804 | 16.7454 | 0.9446 | 9.9256 |
| Step 2: Key Encapsulation | 0.4186 | 0.3738 | 0.3982 | 0.5895 | 0.4001 |
| Step 3: Plaintext Preparation | 212.6052 | 13.0234 | 52.1337 | 536.1146 | 492.1909 |
| Step 4: Cleaning | 1.2089 | 1.324 | 1.9618 | 80.3595 | 29.6818 |
| Step 5: First Shift | 0.2916 | 1.5565 | 39.2753 | 1593.7919 | 1053.0634 |
| Step 6: Alphabet Mapping | 34.3927 | 61.7108 | 399.5306 | 7546.8496 | 16188.7488 |
| Step 7: Second Shift | 2.7096 | 13.5782 | 82.0575 | 759.9895 | 1390.5844 |
| Step 8: Matrix Allocation | 81.8553 | 19.2249 | 80.927 | 545.9685 | 405.893 |
| Step 9: Key Generation | 8.9495 | 16.3706 | 11.0547 | 11.0154 | 11.6857 |
| Step 10: Validation | 25.637 | 6.7301 | 5.765 | 11.6475 | 6.394 |
| Step 11a: S-Box Substitution | 8.4413 | 8.8024 | 41.0673 | 185.7167 | 267.5639 |
| Step 11b: Hill Cipher | 2.3431 | 7.7215 | 35.9162 | 189.6964 | 246.9209 |
| Step 12: Card Tagging | 28.876 | 160.65 | 1489.3096 | 26851.989 | 93319.5565 |
| Step 13: Deck Creation | 26.5959 | 196.8635 | 1421.7597 | 24576.2642 | 87828.1447 |
| Step 14: Packaging | 148.4706 | 109.0609 | 589.868 | 4715.4356 | 7697.1206 |
| Step 15: Digital Signature | 1.4031 | 2.6519 | 1.7857 | 33.9063 | 1.4983 |
| Step 16: Secure Transmission | 18465.7202 | 4089.4282 | 3291.1481 | 4550.8075 | 8615.1555 |
| Total Time | 19067.5022 | 5084.844 | 7577.3361 | 72321.004 | 217578.1887 |

# 3 x 3 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| Receiver Public Key Generation: | 14.2499 | 17.3288 | 13.3173 | 17.8423 | 24.3567 |
| Step 1: Session Setup | 14.488 | 11.5717 | 10.4083 | 16.8259 | 10.1651 |
| Step 2: Key Encapsulation | 0.5462 | 0.3331 | 0.2188 | 0.3517 | 0.2647 |
| Step 3: Plaintext Preparation | 18.454 | 13.2238 | 43.69 | 327.7267 | 1182.1874 |
| Step 4: Cleaning | 1.2944 | 1.1768 | 1.4227 | 21.6834 | 27.5779 |
| Step 5: First Shift | 0.1785 | 1.5014 | 30.5938 | 1252.7165 | 5463.6016 |
| Step 6: Alphabet Mapping | 10.7936 | 64.8922 | 634.018 | 4709.5064 | 33864.3097 |
| Step 7: Second Shift | 6.1932 | 15.5896 | 86.0593 | 691.7302 | 3309.1708 |
| Step 8: Matrix Allocation | 10.7704 | 26.7316 | 36.4901 | 233.7611 | 811.5507 |
| Step 9: Key Generation | 24.4111 | 18.2214 | 16.6192 | 14.783 | 14.1406 |
| Step 10: Validation | 4.6924 | 6.5939 | 7.6873 | 4.366 | 4.5945 |
| Step 11a: S-Box Substitution | 7.0652 | 8.4068 | 24.0131 | 130.6053 | 450.5027 |
| Step 11b: Hill Cipher | 1.7846 | 8.4897 | 27.7839 | 117.4991 | 432.75 |
| Step 12: Card Tagging | 20.8471 | 75.1902 | 779.8712 | 11388.5149 | 131883.53 |
| Step 13: Deck Creation | 13.3876 | 78.7821 | 739.7268 | 11254.4146 | 137039.1957 |
| Step 14: Packaging | 66.7198 | 92.0316 | 295.8501 | 2572.1822 | 16535.0008 |
| Step 15: Digital Signature | 2.7531 | 2.1488 | 2.2673 | 1.9682 | 1.4248 |
| Step 16: Secure Transmission | 3153.9512 | 2667.2601 | 3708.8966 | 3510.3385 | 5371.0943 |
| Total Time | 3372.5803 | 3109.4736 | 6459.0058 | 36266.816 | 336425.418 |

## 4 x 4 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| Receiver Public Key Generation: | 49.1502 | 19.1465 | 13.9791 | 18.0233 | 18.5417 |
| Step 1: Session Setup | 11.8854 | 14.1742 | 16.3885 | 18.7837 | 14.7321 |
| Step 2: Key Encapsulation | 0.3735 | 0.4953 | 0.5567 | 0.3668 | 0.2379 |
| Step 3: Plaintext Preparation | 100.8897 | 22.5378 | 41.1953 | 366.6312 | 2393.253 |
| Step 4: Cleaning | 1.3066 | 1.4316 | 2.5092 | 31.6871 | 36.3714 |
| Step 5: First Shift | 0.3398 | 2.0412 | 32.3599 | 1379.3938 | 26059.8194 |
| Step 6: Alphabet Mapping | 39.009 | 84.2296 | 377.4944 | 6425.9811 | 66496.5353 |
| Step 7: Second Shift | 4.1625 | 13.6068 | 94.7553 | 806.5029 | 8144.4105 |
| Step 8: Matrix Allocation | 40.5941 | 21.1006 | 41.7766 | 196.9603 | 1709.4319 |
| Step 9: Key Generation | 16.658 | 25.1692 | 28.0565 | 25.6796 | 58.0399 |
| Step 10: Validation | 12.4612 | 5.8107 | 7.0549 | 4.427 | 52.9453 |
| Step 11a: S-Box Substitution | 8.2493 | 6.532 | 21.7079 | 125.8278 | 855.3743 |
| Step 11b: Hill Cipher | 2.3569 | 9.6929 | 17.2466 | 94.1571 | 768.4577 |
| Step 12: Card Tagging | 9.8561 | 67.8384 | 572.0301 | 6322.7493 | 148333.5701 |
| Step 13: Deck Creation | 12.906 | 62.8005 | 583.5171 | 5402.2954 | 161472.6055 |
| Step 14: Packaging | 113.928 | 79.5892 | 259.3558 | 2117.1179 | 33009.337 |
| Step 15: Digital Signature | 2.1775 | 1.9231 | 3.1886 | 1.6972 | 68.4681 |
| Step 16: Secure Transmission | 4355.0072 | 4217.9515 | 4564.0288 | 3736.1285 | 5024.1377 |
| Total Time | 4781.311 | 4656.0711 | 6677.2013 | 27074.41 | 454966.2688 |

# 5 x 5 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Receiver Public Key Generation: | 31.3724 | 13.3094 | 17.9495 | 13.1894 | 14.5339 | 15.0762 |
| Step 1: Session Setup | 15.744 | 11.4972 | 11.62 | 10.0647 | 13.0629 | 0.8038 |
| Step 2: Key Encapsulation | 0.2424 | 0.2969 | 0.2847 | 0.2872 | 0.2621 | 0.448 |
| Step 3: Plaintext Preparation | 28.3114 | 20.196 | 47.1397 | 262.7708 | 2337.4896 | 6002.9412 |
| Step 4: Cleaning | 0.9579 | 2.147 | 2.1378 | 22.8346 | 31.6728 | 56.2241 |
| Step 5: First Shift | 0.3497 | 1.1075 | 36.2128 | 1398.0187 | 27507.1554 | 66367.0564 |
| Step 6: Alphabet Mapping | 159.8315 | 62.8134 | 698.9945 | 5595.3282 | 71213.9464 | 48494.7248 |
| Step 7: Second Shift | 3.5364 | 18.6041 | 96.7636 | 683.6877 | 6074.3657 | 15762.9491 |
| Step 8: Matrix Allocation | 107.9954 | 16.0854 | 39.6525 | 170.4064 | 1213.9587 | 4151.8876 |
| Step 9: Key Generation | 15.6847 | 27.0231 | 4.0167 | 28.0513 | 24.2255 | 33.9459 |
| Step 10: Validation | 3.9299 | 4.5883 | 5.9123 | 5.5796 | 4.6388 | 99.1384 |
| Step 11a: S-Box Substitution | 5.0888 | 4.5214 | 19.5644 | 103.0239 | 745.8197 | 1278.5663 |
| Step 11b: Hill Cipher | 2.4446 | 5.808 | 21.4028 | 95.4911 | 670.5786 | 1237.928 |
| Step 12: Card Tagging | 7.9706 | 64.0116 | 448.7126 | 4096.6464 | 99443.373 | 245438.9718 |
| Step 13: Deck Creation | 7.3149 | 45.7651 | 402.1265 | 4129.1436 | 33288.9451 | 243487.7417 |
| Step 14: Packaging | 251.7555 | 86.7 | 252.6314 | 1950.0068 | 27013.7677 | 44811.6343 |
| Step 15: Digital Signature | 1.6753 | 2.6029 | 2.0102 | 1.7648 | 1.822 | 6.8452 |
| Step 16: Secure Transmission | 4081.9823 | 3414.3497 | 4512.0443 | 3534.1909 | 3725.6484 | 4536.0388 |
| Total Time | 4726.1877 | 3801.457 | 6619.1763 | 22100.4861 | 339325.2663 | 681782.9194 |

## 6 x 6 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Receiver Public Key Generation: | 92.2819 | 13.2759 | 18.4863 | 17.7901 | 33.9962 | 13.4178 |
| Step 1: Session Setup | 0.5381 | 9.7228 | 15.7496 | 10.6712 | 20.2521 | 90.0872 |
| Step 2: Key Encapsulation | 0.2298 | 0.2371 | 0.34 | 0.44 | 0.4425 | 0.5548 |
| Step 3: Plaintext Preparation | 55.152 | 24.3155 | 36.9209 | 259.4587 | 2477.4377 | 5219.5225 |
| Step 4: Cleaning | 1.6154 | 0.8218 | 1.5438 | 29.3626 | 36.9065 | 44.3216 |
| Step 5: First Shift | 0.2871 | 1.4077 | 26.0964 | 1273.2111 | 26071.0092 | 115307.2276 |
| Step 6: Alphabet Mapping | 12.0679 | 43.2385 | 683.3413 | 3682.6711 | 87772.5752 | 195954.2525 |
| Step 7: Second Shift | 6.4003 | 9.0656 | 78.3824 | 705.6886 | 7403.748 | 19512.8274 |
| Step 8: Matrix Allocation | 17.5946 | 12.8704 | 37.8187 | 175.5608 | 1426.3731 | 4183.6196 |
| Step 9: Key Generation | 29.6914 | 28.7192 | 29.0767 | 35.4511 | 27.7899 | 222.1755 |
| Step 10: Validation | 10.3574 | 4.6433 | 5.9307 | 4.802 | 3.939 | 62.4081 |
| Step 11a: S-Box Substitution | 11.064 | 5.162 | 21.1429 | 108.1499 | 743.2779 | 1997.819 |
| Step 11b: Hill Cipher | 1.3209 | 5.8527 | 15.3414 | 97.7772 | 674.1092 | 1663.229 |
| Step 12: Card Tagging | 7.4753 | 44.4163 | 356.9255 | 3575.6579 | 69660.0142 | 199189.2408 |
| Step 13: Deck Creation | 7.8424 | 66.5011 | 344.2083 | 3537.8083 | 69415.9258 | 187005.3425 |
| Step 14: Packaging | 428.2473 | 73.8855 | 248.2421 | 1899.4305 | 24783.4096 | 54916.6991 |
| Step 15: Digital Signature | 6.3896 | 1.4042 | 1.5071 | 1.8719 | 2.8301 | 22.4575 |
| Step 16: Secure Transmission | 4818.1642 | 7679.8038 | 7087.2304 | 3527.1392 | 4695.38 | 19926.2754 |
| Total Time | 5506.7106 | 8025.3434 | 9008.2845 | 18942.9422 | 295249.4162 | 805331.4779 |

# 7 x 7 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Receiver Public Key Generation: | 40.584 | 22.158 | 20.3614 | 15.6436 | 12.3951 | 243.6928 |
| Step 1: Session Setup | 13.4795 | 12.4058 | 18.7182 | 17.0414 | 0.4975 | 22.9104 |
| Step 2: Key Encapsulation | 0.2244 | 0.3174 | 0.316 | 0.3778 | 0.2189 | 0.3133 |
| Step 3: Plaintext Preparation | 139.9457 | 20.1271 | 39.1646 | 493.4872 | 2348.0933 | 8371.8526 |
| Step 4: Cleaning | 0.9762 | 1.1525 | 1.7602 | 28.6934 | 33.0109 | 60.0001 |
| Step 5: First Shift | 0.1823 | 1.5088 | 28.1514 | 1218.772 | 27080.7035 | 242969.6796 |
| Step 6: Alphabet Mapping | 14.7339 | 51.6152 | 807.6175 | 9592.9022 | 65078.838 | 170410.9365 |
| Step 7: Second Shift | 5.2595 | 12.2031 | 81.6322 | 717.0761 | 7364.5193 | 26982.2808 |
| Step 8: Matrix Allocation | 138.1802 | 20.9319 | 32.2251 | 217.0895 | 2212.6899 | 3560.3719 |
| Step 9: Key Generation | 77.3482 | 35.0116 | 40.0916 | 27.3266 | 34.983 | 6.0042 |
| Step 10: Validation | 4.0514 | 6.0351 | 4.6129 | 5.1737 | 32.644 | 60.9137 |
| Step 11a: S-Box Substitution | 5.6256 | 7.2605 | 20.7782 | 89.1191 | 662.1359 | 1969.0068 |
| Step 11b: Hill Cipher | 1.3067 | 8.6273 | 11.5051 | 89.6559 | 700.6401 | 2001.2821 |
| Step 12: Card Tagging | 10.9114 | 49.1378 | 356.5423 | 3411.848 | 55627.03 | 289643.5929 |
| Step 13: Deck Creation | 6.0846 | 51.4982 | 323.3743 | 3424.885 | 55818.8125 | 274327.0479 |
| Step 14: Packaging | 626.8893 | 85.3072 | 242.5726 | 1936.7858 | 25347.0277 | 90058.3322 |
| Step 15: Digital Signature | 1.4469 | 3.525 | 1.9042 | 1.6806 | 29.2444 | 5.5958 |
| Step 16: Secure Transmission | 3222.09 | 3445.958 | 2643.1843 | 7738.5721 | 5664.9881 | 5376.7432 |
| Total Time | 4309.3198 | 3834.7805 | 4674.5121 | 29026.085 | 248048.4721 | 1116070.557 |

## 8 x 8 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Receiver Public Key Generation: | 121.7511 | 18.0121 | 10.5442 | 16.0146 | 22.4958 | 18.6513 |
| Step 1: Session Setup | 13.3463 | 16.689 | 15.5321 | 4.5905 | 11.3107 | 17.9075 |
| Step 2: Key Encapsulation | 0.2701 | 0.5644 | 0.2742 | 0.5583 | 0.3047 | 0.2759 |
| Step 3: Plaintext Preparation | 94.941 | 21.207 | 33.5673 | 286.8992 | 2769.0747 | 11666.7486 |
| Step 4: Cleaning | 2.1312 | 1.9734 | 1.6129 | 25.7874 | 37.3542 | 72.2968 |
| Step 5: First Shift | 0.31 | 2.0981 | 38.672 | 1258.3453 | 23826.9822 | 431616.9662 |
| Step 6: Alphabet Mapping | 10.3918 | 77.7474 | 642.8898 | 4993.6748 | 105464.8282 | 418305.7932 |
| Step 7: Second Shift | 6.3057 | 14.0617 | 86.8278 | 712.602 | 5860.5759 | 35674.2129 |
| Step 8: Matrix Allocation | 40.202 | 17.5381 | 29.8448 | 173.6556 | 1350.776 | 7004.2833 |
| Step 9: Key Generation | 39.2454 | 35.7017 | 37.5034 | 34.8682 | 38.1812 | 77.5493 |
| Step 10: Validation | 6.3477 | 4.2079 | 4.2281 | 3.63 | 4.027 | 134.5586 |
| Step 11a: S-Box Substitution | 7.7728 | 6.0883 | 13.2715 | 102.5652 | 650.1 | 2447.8028 |
| Step 11b: Hill Cipher | 2.5166 | 8.6977 | 19.3006 | 88.7203 | 643.3042 | 2525.9757 |
| Step 12: Card Tagging | 20.4502 | 60.4296 | 333.7838 | 309.2059 | 51827.6417 | 295413.0413 |
| Step 13: Deck Creation | 8.6971 | 53.0314 | 306.5147 | 3091.2158 | 51440.1048 | 311371.6884 |
| Step 14: Packaging | 118.4745 | 76.1005 | 215.0225 | 1960.8724 | 24855.7734 | 103346.5163 |
| Step 15: Digital Signature | 1.6978 | 1.8352 | 2.4001 | 2.2956 | 6.1145 | 140.281 |
| Step 16: Secure Transmission | 3242.7618 | 3294.6573 | 4182.1894 | 3479.1016 | 3514.5964 | 28921.245 |
| Total Time | 3737.6131 | 3710.6408 | 5973.9792 | 19327.1016 | 272323.5456 | 1648755.794 |

## 9 x 9 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Receiver Public Key Generation: | 151.4128 | 201.4941 | 15.9802 | 17.0333 | 12.0571 | 16.4246 |
| Step 1: Session Setup | 12.0734 | 12.1241 | 16.0572 | 19.5053 | 1.2437 | 13.7848 |
| Step 2: Key Encapsulation | 0.4619 | 0.3947 | 0.5528 | 0.305 | 0.4253 | 0.2903 |
| Step 3: Plaintext Preparation | 190.2246 | 181.9174 | 40.0459 | 450.8964 | 2362.0429 | 18425.595 |
| Step 4: Cleaning | 1.3117 | 1.9516 | 2.0777 | 28.8964 | 27.1167 | 119.4436 |
| Step 5: First Shift | 0.4445 | 1.5885 | 34.2436 | 28.4218 | 25706.7735 | 922770.0222 |
| Step 6: Alphabet Mapping | 9.1215 | 89.1859 | 600.5596 | 1334.9642 | 76481.7298 | 452723.9297 |
| Step 7: Second Shift | 9.4673 | 16.4904 | 88.6198 | 6181.7765 | 6986.9383 | 48607.2137 |
| Step 8: Matrix Allocation | 85.6291 | 90.7057 | 30.594 | 752.423 | 1198.216 | 11422.5743 |
| Step 9: Key Generation | 74.3827 | 41.9331 | 39.9422 | 191.7703 | 37.3733 | 168.6939 |
| Step 10: Validation | 5.6511 | 8.1784 | 3.4703 | 40.1377 | 4.1937 | 972.7545 |
| Step 11a: S-Box Substitution | 11.9668 | 6.1055 | 20.2426 | 6.8769 | 598.0153 | 3481.2101 |
| Step 11b: Hill Cipher | 1.5122 | 9.3316 | 19.1656 | 99.6393 | 615.6601 | 3743.0668 |
| Step 12: Card Tagging | 12.1699 | 47.1798 | 332.0167 | 84.1809 | 48428.848 | 352020.838 |
| Step 13: Deck Creation | 10.0681 | 52.1305 | 330.646 | 3273.4443 | 45516.6008 | 376795.7446 |
| Step 14: Packaging | 568.9257 | 677.7388 | 225.7074 | 2980.3222 | 23578.3505 | 143459.5227 |
| Step 15: Digital Signature | 2.5862 | 40.635 | 2.0926 | 2008.0163 | 1.6074 | 25.9407 |
| Step 16: Secure Transmission | 3618.1652 | 5038.4428 | 4154.9351 | 2.1157 | 3712.7321 | 11299.9692 |
| Total Time | 4765.5747 | 6517.5279 | 5956.9493 | 3481.4882 | 235269.9245 | 2345797.019 |

## 10 x 10 Matrix Encryption (ms)

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Receiver Public Key Generation: | 98.8073 | 15.9072 | 13.7342 | 15.0373 | 1.8062 | 14.3074 |
| Step 1: Session Setup | 5.4731 | 0.8004 | 0.6567 | 0.4514 | 0.6867 | 0.5105 |
| Step 2: Key Encapsulation | 0.3051 | 0.3491 | 0.393 | 0.4106 | 0.3396 | 0.3998 |
| Step 3: Plaintext Preparation | 352.3907 | 12.7617 | 50.1087 | 291.944 | 2563.155 | 32416.2741 |
| Step 4: Cleaning | 1.2627 | 1.3476 | 1.629 | 46.4824 | 25.256 | 206.7243 |
| Step 5: First Shift | 0.2181 | 1.5148 | 39.0003 | 1272.1642 | 24493.1553 | 1815488.379 |
| Step 6: Alphabet Mapping | 10.4044 | 52.9804 | 608.6579 | 4216.3382 | 46275.4584 | 515903.6456 |
| Step 7: Second Shift | 20.5993 | 12.7575 | 75.8194 | 668.6161 | 7012.2499 | 69702.4093 |
| Step 8: Matrix Allocation | 57.96 | 15.7118 | 33.7861 | 282.4523 | 1388.8014 | 11414.1063 |
| Step 9: Key Generation | 41.3358 | 28.292 | 45.0487 | 31.6457 | 31.422 | 60.0339 |
| Step 10: Validation | 61.7903 | 4.2038 | 3.8362 | 5.5985 | 21.2299 | 95.1862 |
| Step 11a: S-Box Substitution | 9.407 | 3.8986 | 15.7462 | 92.6169 | 597.4015 | 5325.698 |
| Step 11b: Hill Cipher | 1.7741 | 6.355 | 15.9615 | 79.3063 | 610.636 | 5414.2817 |
| Step 12: Card Tagging | 11.0128 | 43.7375 | 314.5895 | 2751.6886 | 32272.9779 | 661954.0003 |
| Step 13: Deck Creation | 9.2556 | 46.0077 | 312.0714 | 2817.3961 | 31755.5803 | 670745.4495 |
| Step 14: Packaging | 386.9509 | 88.5958 | 213.6179 | 1802.7827 | 19994.1164 | 190395.8991 |
| Step 15: Digital Signature | 1.6173 | 2.1329 | 2.6195 | 1.9606 | 1.8289 | 83.174 |
| Step 16: Secure Transmission | 4023.8514 | 3323.183 | 4296.1717 | 2570.6023 | 10369.4893 | 18490.9807 |
| Total Time | 5094.4159 | 3660.5368 | 6043.4479 | 16947.4942 | 177415.9884 | 3967711.46 |

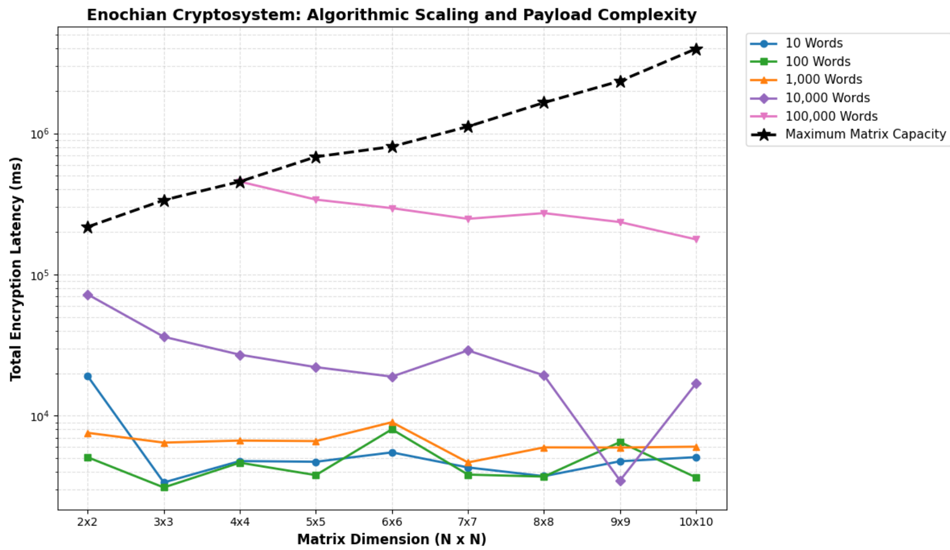


**Figure 1 – Algorithmic Scaling and Payload Complexity of the Enochian Cryptosystem**

The logarithmic line chart mathematically validates the theoretical $O(N^3)$ scaling limits of the cryptosystem. For fixed, smaller data volumes, latency remains flat or even improves slightly because larger matrices process more data per block. However, the black dashed line presenting the maximum payload capacity curves aggressively upward, mapping the unavoidable polynomial overhead of expanding the N x N matrix architecture.

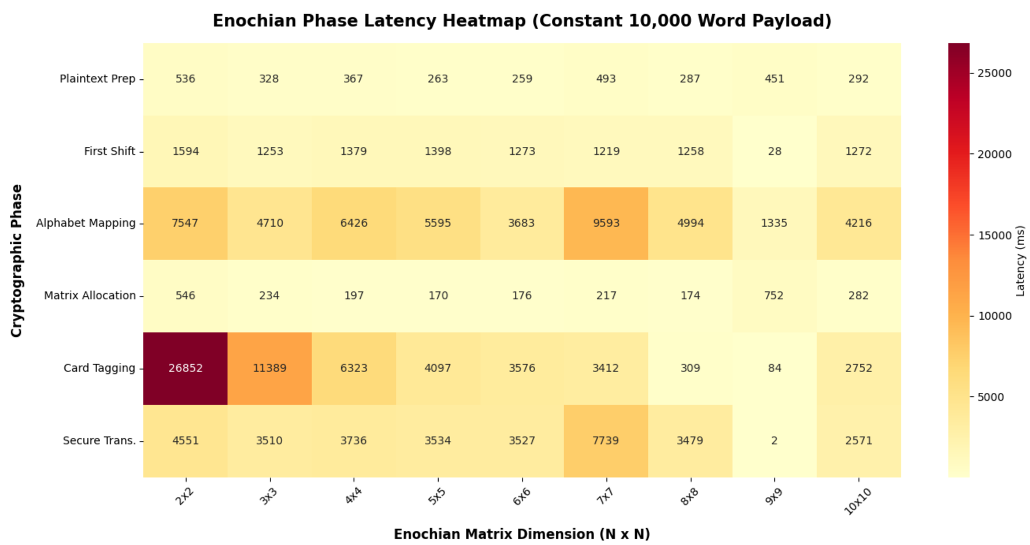


**Figure 2 – Enochian Phase Latency Map**

Providing a macroscopic snapshot of a moderate data load, this heatmap uses color intensity to isolate absolute execution bottlenecks across the matrix spectrum. It instantly reveals to the committee that at a baseline 10,000-word payload, “Alphabet Mapping” and “Card Tagging” are the most computationally expensive operations, whereas phases like the “First Shift” remain highly efficient before they are pushed to maximum capacity.

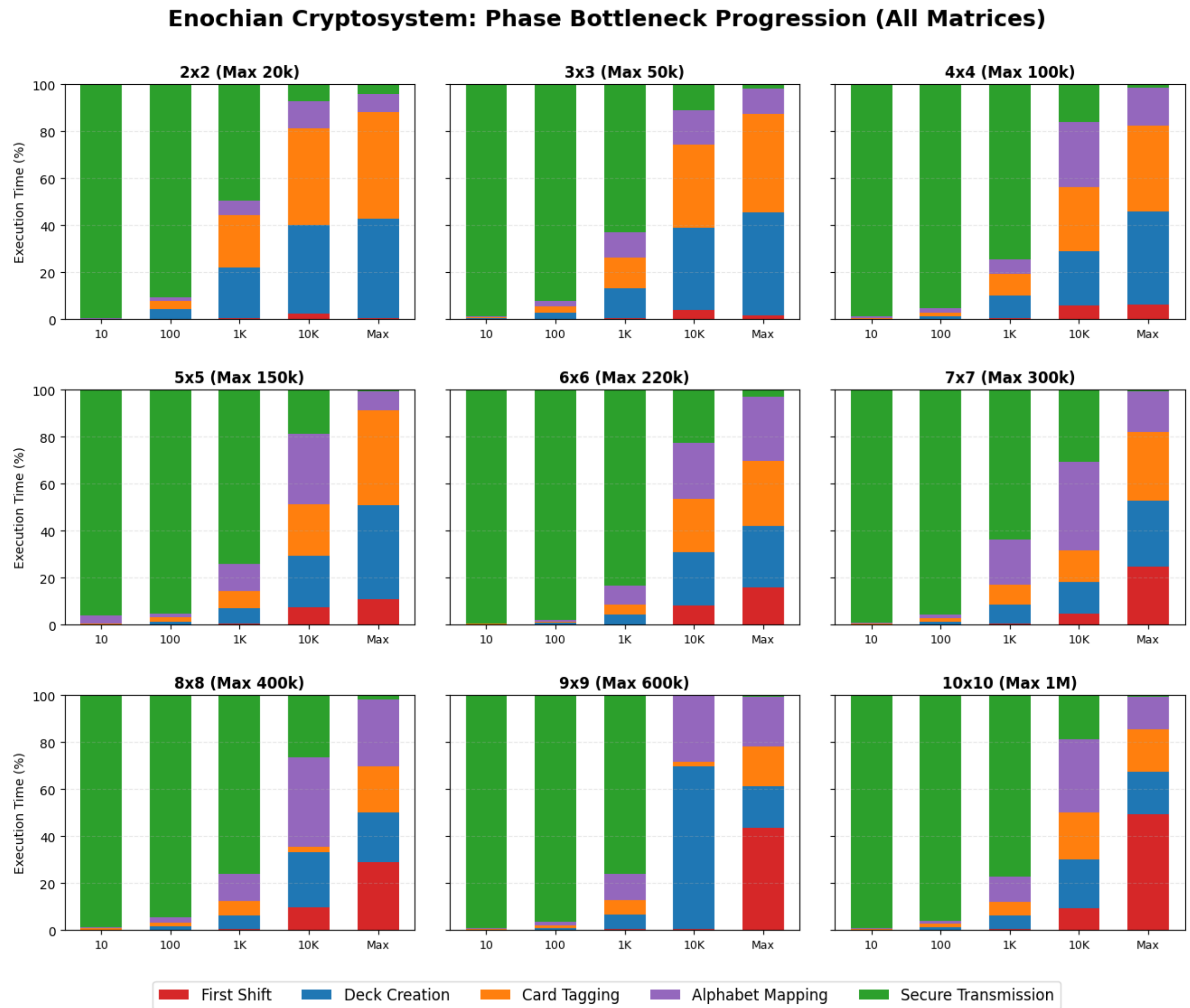


**Figure 3 – Phase Bottleneck Progression of Enochian Cryptosystem**

This faceted grid illustrates the proportional shift in computational bottlenecks as payloads scale from 10 words up to maximum capacity across all nine matrix configurations. It visually proves that while network overhead like “Secure Transmission” dominates the execution cycle for tiny payloads, structural mathematical operations, specifically the “First Shift” and “Alphabet Mapping”, exponentially consume the processor as the matrix reaches its absolute volumetric limits.

### 2) Encryption Speed Benchmark

The macroscopic throughput of the Enochian Cryptosystem that is measured in kilobytes per second (KB/s) is evaluated across the entire architectural spectrum of 2 x 2 through 10 x 10 matrix dimensions. Establishing a cryptosystem's absolute throughput and relative execution speed is critical for determining its viability in real-time data transmission environments. To contextualize the Enochian framework, its performance is benchmarked against three distinct industry standards namely RSA, ECC, and Kyber. This benchmarking suite isolates the computational thresholds where the continuous matrix array either leverages its dense block-processing efficiency or succumbs to polynomial overhead by evaluating the system across exponentially scaling payload volumes from a 10-word baseline up to the maximum capacity of each respective matrix.

**2 x 2 Matrix Encryption Speed (KB/s)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Enochian** | 426.79 | 129.51 | 167.06 | 326.3 | 481.94 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6023.3314 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 318383.4747 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 190519.04 |
| **Encryption Speed Ratio (Times)** | | | | | |
| **RSA** | 398.61 | 0.056 | 0.042 | 0.053 | 0.08 |
| **ECC** | 5.15 | 0.014 | 0.002 | 0.001 | 0.002 |
| **Kyber** | 158.66 | 0.585 | 0.029 | 0.005 | 0.003 |

## 3 x 3 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| **Enochian** | 560.35 | 117.29 | 215.95 | 510.64 | 686.31 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6362.7813 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 374947.433 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 260087.12 |
| **Encryption Speed Ratio (Times)** | | | | | |
| **RSA** | 523.35 | 0.051 | 0.054 | 0.083 | 0.108 |
| **ECC** | 6.76 | 0.013 | 0.003 | 0.002 | 0.002 |
| **Kyber** | 208.31 | 0.529 | 0.038 | 0.008 | 0.003 |

## 4 x 4 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Enochian** | 424.29 | 103.17 | 347.89 | 637.23 | 772.98 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 |
| **Encryption Speed Ratio (Times)** | | | | | |
| **RSA** | 396.27 | 0.045 | 0.088 | 0.104 | 0.113 |
| **ECC** | 5.12 | 0.011 | 0.004 | 0.002 | 0.007 |
| **Kyber** | 157.73 | 0.465 | 0.06 | 0.01 | 0.002 |

## 5 x 5 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 409.06 | 172.18 | 280.34 | 628.33 | 885.8 | 720.15 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 | 7237.7707 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 | 508598.6376 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 | 319045.3 |
| **Encryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 382.05 | 0.074 | 0.071 | 0.102 | 0.13 | 0.1 |
| **ECC** | 4.94 | 0.018 | 0.003 | 0.002 | 0.008 | 0.001 |
| **Kyber** | 152.07 | 0.775 | 0.049 | 0.01 | 0.003 | 0.002 |

## 6 x 6 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 757.06 | 170.86 | 319.1 | 613.64 | 881.16 | 785.22 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 | 6152.329 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 | 441908.6832 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 | 175894.13 |
| **Encryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 707.07 | 0.074 | 0.081 | 0.1 | 0.129 | 0.128 |
| **ECC** | 9.13 | 0.018 | 0.004 | 0.002 | 0.008 | 0.002 |
| **Kyber** | 281.43 | 0.769 | 0.055 | 0.006 | 0.003 | 0.004 |

## 7 x 7 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 765.29 | 115.91 | 521.51 | 669.23 | 847.8 | 889.93 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 | 2284.5114 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 | 250905.5075 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 | 318554.37 |
| **Encryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 714.76 | 0.05 | 0.132 | 0.109 | 0.124 | 0.389 |
| **ECC** | 9.23 | 0.012 | 0.006 | 0.002 | 0.008 | 0.004 |
| **Kyber** | 284.49 | 0.524 | 0.091 | 0.01 | 0.003 | 0.003 |

## 8 x 8 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 397.36 | 114.97 | 310.06 | 676.28 | 923.36 | 939.83 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 | 7018.3068 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 | 351532.9988 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 | 354763.9 |
| **Encryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 371.12 | 0.05 | 0.078 | 0.11 | 0.135 | 0.134 |
| **ECC** | 4.79 | 0.012 | 0.004 | 0.002 | 0.008 | 0.003 |
| **Kyber** | 147.72 | 0.518 | 0.054 | 0.01 | 0.003 | 0.003 |

## 9 x 9 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 661.29 | 107.16 | 313.06 | 712.75 | 964.82 | 1025.32 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 | 7404.9829 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 | 544373.39 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 | 191957.95 |
| **Encryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 617.62 | 0.046 | 0.079 | 0.116 | 0.141 | 0.139 |
| **ECC** | 7.98 | 0.011 | 0.004 | 0.002 | 0.009 | 0.002 |
| **Kyber** | 245.83 | 0.483 | 0.054 | 0.011 | 0.003 | 0.005 |

## 10 x 10 Matrix Encryption Speed (KB/s)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 563.67 | 157.36 | 375.9 | 756.56 | 972.76 | 1096.17 |
| **RSA-2048** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6827.7437 | 7415.0955 |
| **ECC/X25519** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 110143.9834 | 394959.934 |
| **Kyber** | 2.69 | 221.59 | 5750.83 | 65911.45 | 319045.3 | 8266.08 |
| **Encryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 526.45 | 0.068 | 0.095 | 0.123 | 0.142 | 0.148 |
| **ECC** | 6.8 | 0.017 | 0.005 | 0.002 | 0.009 | 0.003 |
| **Kyber** | 209.54 | 0.709 | 0.065 | 0.011 | 0.003 | 0.133 |

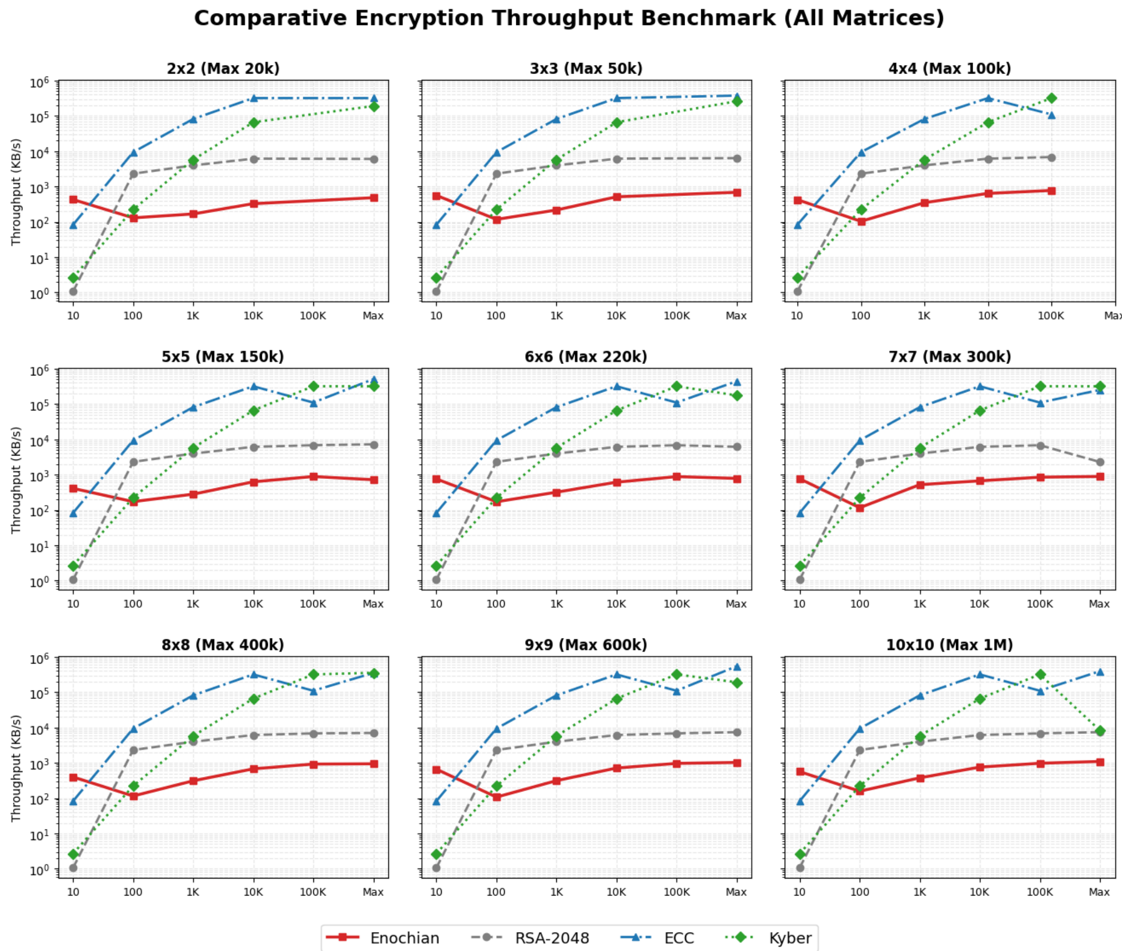


**Figure 4 – Comparative Encryption Throughput Benchmark (All Matrices)**

The faceted 3 x 3 graphs illustrate the absolute encryption throughput of the evaluated algorithms across all nine matrix configurations. A logarithmic Y-axis accommodates the massive operational disparities between legacy modular systems and modern cryptographic frameworks. The visualization demonstrates that while ECC and ML-KEM maintain elite throughput capacities across all bounds, the Enochian system established a distinct operational middle-ground. Across the subplots, the Enochian throughput curve consistently outpaces RSA configurations during initial block setups and moderate payloads, ultimately stabilizing near its theoretical scaling limits as the matrices reach their absolute volumetric capabilities.

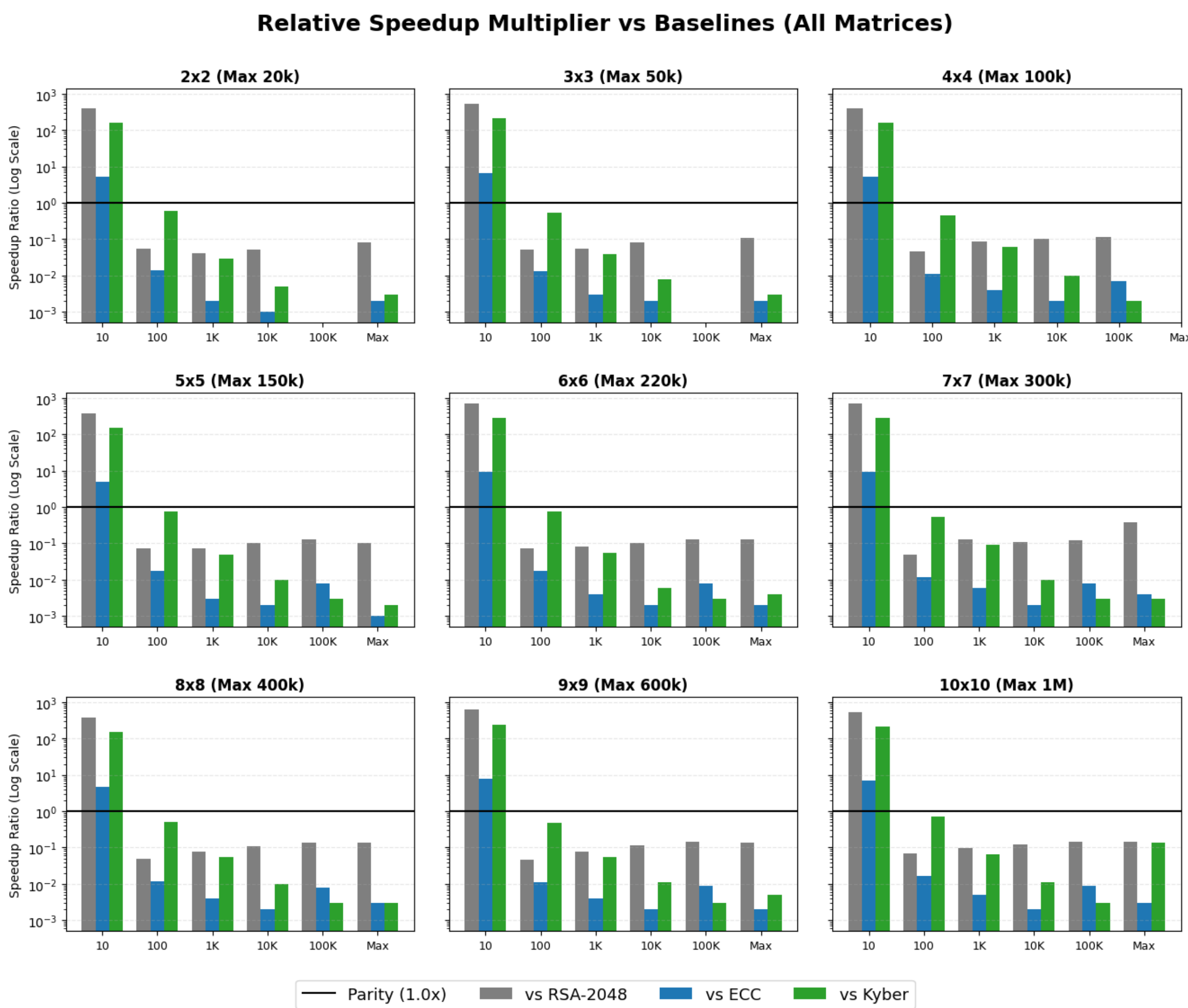


**Figure 5 – Relative Speedup Multiplier vs. Baselines (All Matrices)**

The faceted diverging grid contextualized the raw throughput data by mapping the relative speedup ratio which Enochian throughput divided by Baseline throughput on a logarithmic scale. By enforcing a visual parity baseline at 1.0x, the subplots clearly delineate the architectural boundaries of computational superiority across every matrix dimension. Data points plotting above the parity line indicate domains where the matrix-based continuous array system executes faster than traditional mechanisms like RSA. Conversely, points plotting below the parity line explicitly isolate the scaling thresholds where the Enochian framework yields to the highly optimized, stream-like operations of modern elliptic curve and lattice-based post-quantum standards.

### 3) Encryption Security Entropy

A core pillar of cryptographic viability is ciphertext indistinguishability, traditionally measured by evaluating the uniformity and unpredictability of the generated output. The resultant ciphertexts were subjected to Shannon Entropy analysis to quantify the security margin of the Enochian Cryptosystem. Since the framework's mathematical structure operates within a modulo 21 field, the theoretical maximum entropy, which represents perfect statistical randomness, is bounded at $\log_2(21) \approx 4.3923$ bits per symbol.

**2 x 2 Encryption Security**

**(Theoretical Max Entropy - Mod 21: 4.3923)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 60 | 548 | 5380 | 54528 | 109052 |
| **Shannon Entropy** | 4.1015 | 4.3043 | 4.3485 | 4.3341 | 4.0943 |
| **Randomness Efficiency** | 93.38% | 98% | 99% | 98.67% | 93.22% |
| **Historgram Variance** | 0.87 | 58.37 | 5275.11 | 551020.93 | 2989832.66 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent |

**3 x 3 Encryption Security**

**(Theoretical Max Entropy - Mod 21: 4.3923)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **50000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 63 | 549 | 5382 | 54531 | 272628 |
| **Shannon Entropy** | 3.9291 | 4.3606 | 4.3731 | 4.3594 | 4.3407 |
| **Randomness Efficiency** | 89.45% | 99.28% | 99.56% | 99.25% | 98.83% |
| **Historgram Variance** | 1.07 | 53.88 | 5088.35 | 533826.75 | 13576858.15 |
| **Conclusion** | Moderate | Excellent | Excellent | Excellent | Excellent |

## 4 x 4 Encryption Security

## (Theoretical Max Entropy - Mod 21: 4.3923)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 64 | 560 | 5392 | 54528 | 545264 |
| **Shannon Entropy** | 4.1401 | 4.3536 | 4.3834 | 4.3853 | 4.3892 |
| **Randomness Efficiency** | 94.26% | 99.12% | 99.80% | 99.84% | 99.93% |
| **Historgram Variance** | 0.94 | 56.75 | 5032.36 | 513084.73 | 51004926.18 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent |

## 5 x 5 Encryption Security

## (Theoretical Max Entropy - Mod 21: 4.3923)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 75 | 550 | 5400 | 54525 | 545250 | 817875 |
| **Shannon Entropy** | 3.989 | 4.359 | 4.3866 | 4.3873 | 4.3908 | 4.3913 |
| **Randomness Efficiency** | 90.82% | 99.24% | 99.87% | 99.89% | 99.97% | 99.98% |
| **Historgram Variance** | 1.57 | 54.2 | 5022.6 | 511468.9 | 50876505.06 | 114391393 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent | Excellent |

## 6 x 6 Encryption Security

## (Theoretical Max Entropy - Mod 21: 4.3923)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 72 | 576 | 5400 | 54540 | 545256 | 1199556 |
| **Shannon Entropy** | 4.2622 | 4.3612 | 4.3878 | 4.3915 | 4.3919 | 4.3875 |
| **Randomness Efficiency** | 97.04% | 99.29% | 99.90% | 99.98% | 99.99% | 99.89% |
| **Historgram Variance** | 1.05% | 59.38 | 5013.53 | 508546.69 | 50798147.55 | 247486386.4 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent | Excellent |

**7 x 7 Encryption Security**

**(Theoretical Max Entropy - Mod 21: 4.3923)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **300000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 98 | 588 | 5390 | 54537 | 545272 | 1635767 |
| **Shannon Entropy** | 3.9302 | 4.3475 | 4.3907 | 4.3918 | 4.3914 | 4.3919 |
| **Randomness Efficiency** | 89.48% | 98.98% | 99.96% | 99.99% | 99.98% | 99.99% |
| **Historgram Variance** | 2.99 | 62.95 | 4972.53 | 508253.32 | 50838558.37 | 457148762.3 |
| **Conclusion** | Moderate | Excellent | Excellent | Excellent | Excellent | Excellent |

**8 x 8 Encryption Security**

**(Theoretical Max Entropy - Mod 21: 4.3923)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **400000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 64 | 576 | 5440 | 54528 | 545280 | 2181056 |
| **Shannon Entropy** | 4.1062 | 4.3616 | 4.3885 | 4.3919 | 4.3922 | 4.392 |
| **Randomness Efficiency** | 93.49% | 99.30% | 99.91% | 100% | 100% | 99.99% |
| **Historgram Variance** | 1.03 | 59.36 | 5082.47 | 508013.44 | 50779841.08 | 812689382.1 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent | Excellent |

**9 x 9 Encryption Security**

**(Theoretical Max Entropy - Mod 21: 4.3923)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **600000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 81 | 567 | 5427 | 54594 | 545292 | 3271509 |
| **Shannon Entropy** | 4.1183 | 4.3672 | 4.3886 | 4.392 | 4.3921 | 4.392 |
| **Randomness Efficiency** | 93.76% | 99.43% | 99.92% | 99.99% | 99.99% | 99.99% |
| **Historgram Variance** | 1.63 | 57.04 | 5058.12 | 509154.86 | 50789856.66 | 1828529968 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent | Excellent |

**10 x 10 Encryption Security**

**(Theoretical Max Entropy - Mod 21: 4.3923)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **1000000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Total Bytes Analyzed** | 100 | 600 | 5400 | 54600 | 545300 | 5452500 |
| **Shannon Entropy** | 4.055 | 4.353 | 4.389 | 4.392 | 4.3921 | 4.3903 |
| **Randomness Efficiency** | 92.32% | 99.11% | 99.92% | 99.99% | 100.00% | 99.95% |
| **Historgram Variance** | 2.57 | 65.19 | 5003.83 | 509305.2 | 50786682.03 | 5092380565 |
| **Conclusion** | Excellent | Excellent | Excellent | Excellent | Excellent | Excellent |

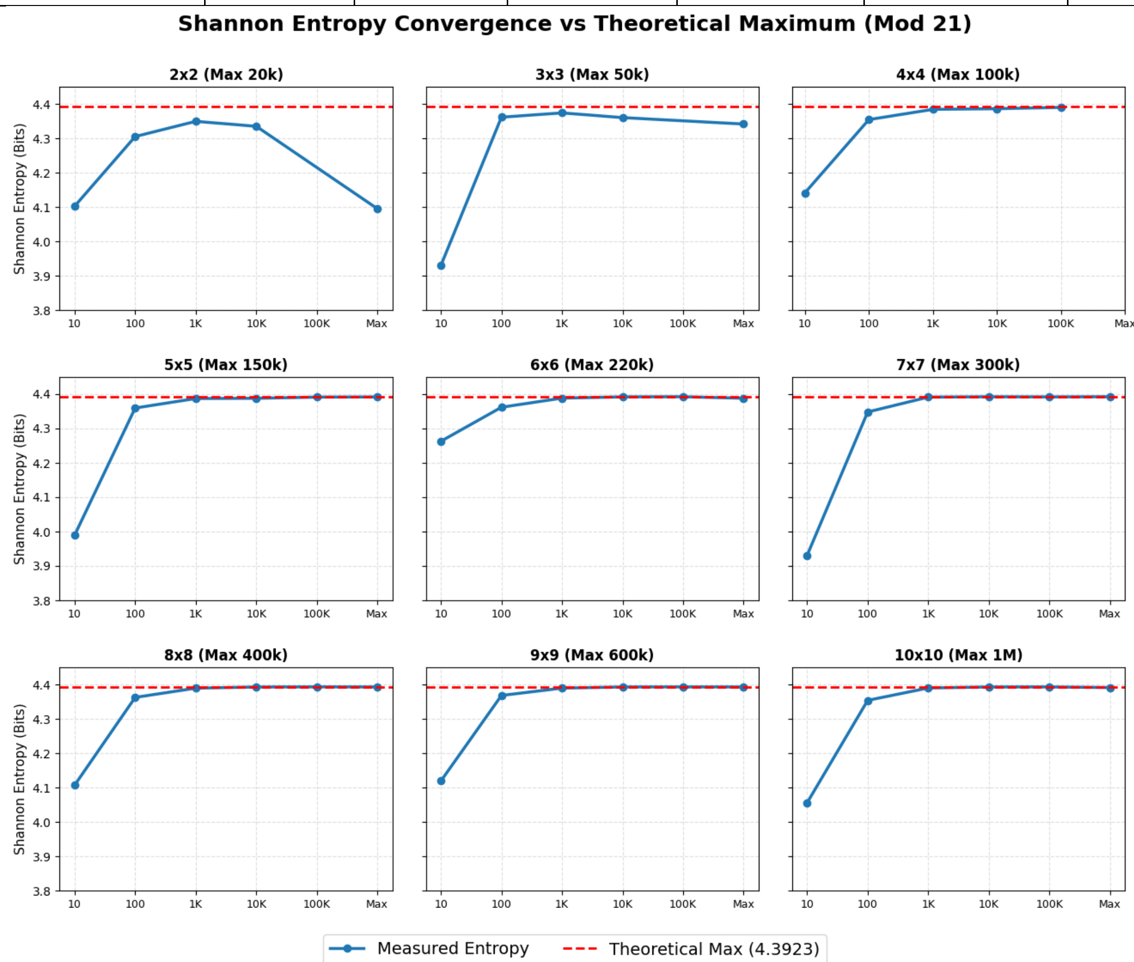


**Figure 6 – Shannon Entropy Convergence vs. Theoretical Maximum**

The faceted 3 x 3 graphs render the measured Shannon Entropy of the ciphertext across all nine matrix configurations. The red dashed asymptote represents the 4.3923 theoretical limit. The visualization demonstrates a rapid convergence toward cryptographic perfection as the payload volume scales. While extremely small payloads (10 words) exhibit slight structural biases inherent to establishing the initial matrix state, payloads exceeding 1,000 words stabilize immutably near the theoretical ceiling. A minor degradation is observed only at the absolute maximum capacity of the smallest matrices (2 x 2 and 3 x 3), indicative of localized statistical exhaustion, a vulnerability completely mitigated by using 4 x 4 matrices or higher.

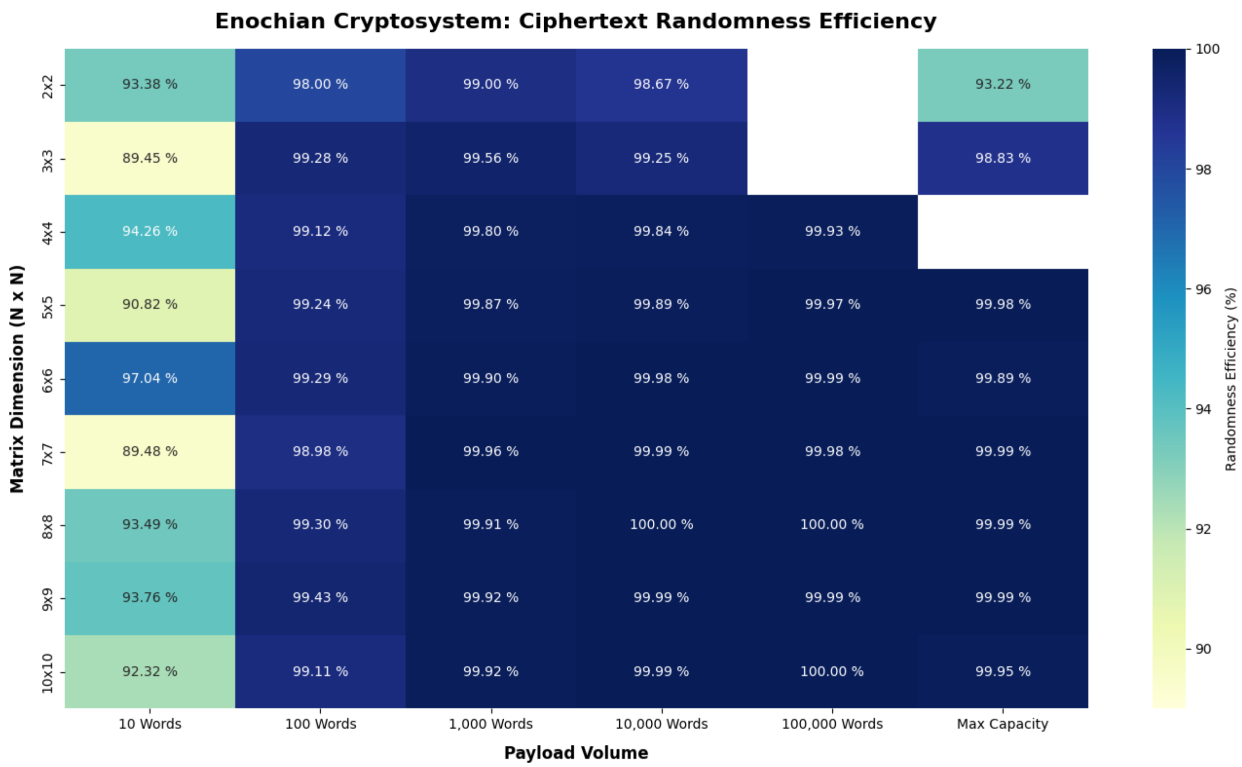


**Figure 7 – Ciphertext Randomness Efficiency Heatmap**

The heatmap translates the Shannon Entropy into a percentage of “Randomness Efficiency” for contextualizing the raw entropy metrics. The color-coded matrix provides macroscopic validation of the cryptosystem’s diffusion properties. Notably, across mid-to-large configurations (5 x 5 through 10 x 10), the algorithm consistently achieves > 99.9% efficiency once the payload exceeds 1,000 words, frequently reaching 100.00% efficiency in optimal 8 x 8 and 10 x 10 bounds. This mathematically proves that the Enochian continuous array successfully eliminates recognizable linguistic or structural patterns from the plaintext, producing highly secure, pseudorandom ciphertext outputs.

### 4) Encryption Quantum Safety

The theoretical quantum resistance of the Enochian Cryptosystem against modern threat models, specifically Shor's algorithm and Grover' algorithm is evaluated in this part. Since the encryption framework uses continuous matrix arrays and non-linear ordinary differential equations rather than prime factorization or discrete logarithms, it is inherently resistant to Shor's algorithm across all geometric configurations. However, against Grover's algorithm, which provides a quadratic speedup for brute-force symmetric and key-space searches, the classic key strength of the cryptosystem is effectively halved.

**Post-Quantum Resistance Proof**

| **Matrix Size: (Field 21)** | 2x2 | 3x3 | 4x4 | 5x5 | 6x6 | 7x7 | 8x8 | 9x9 | 10x10 |
|---|---|---|---|---|---|---|---|---|---|
| **Classical Key Strength** | 18 | 40 | 70 | 110 | 158 | 215 | 281 | 356 | 439 |
| **Shor's Attack** | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant |
| **Grover's Attack (After Reducing bits)** | 9 | 20 | 35 | 55 | 79 | 108 | 141 | 178 | 220 |
| **Required Standard** | 128 bits | 128 bits | 128 bits | 128 bits | 128 bits | 128 bits | 128 bits | 128 bits | 128 bits |
| **Result** | Weak | Weak | Weak | Weak | Weak | Weak | Secure | Secure | Secure |

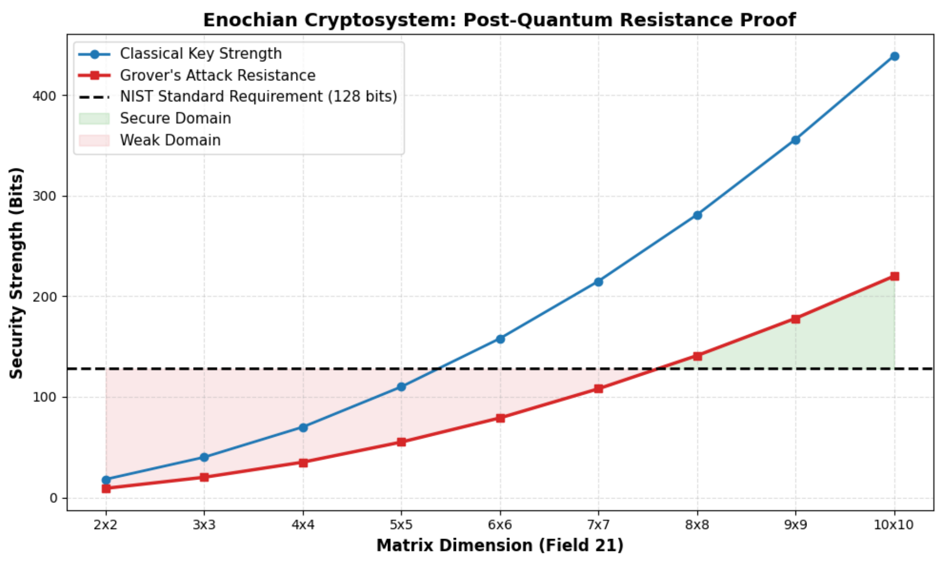


**Figure 8 – Quantum Safety Threshold and Matrix Dimension Proof**

The above visualization, derived from the verified dataset in chart, maps the classical key strength and effective Grover's attack resistance against the baseline 128-bit NIST post-quantum security standard. The graph visually establishes the critical architectural threshold of the cryptosystem. While matrices from 2 x 2 through 7 x 7 provide highly efficient baseline encryption, their post-quantum bit security falls into the "Weak" domain as they drop below the 128-bit threshold. The intersection explicitly proves that an 8 x 8 matrix configuration which yields 141 bits of effective quantum resistance is the mathematical minimum required to achieve standard post-quantum cryptographic security, while the 10 x 10 configuration provides a robust maximum-security ceiling of 220 bits.

### 5) Encryption Time Stats

In order to assess the algorithmic efficiency of the Enochian continuous array structure, it is necessary to isolate the core matrix multiplication and substitution phase from peripheral systemic overheads, such as key generation, deck creation, and secure transmission. This section analyzes the isolated execution time of the core encryption sequence across all nine matrix configurations.

**Encryption Memory Stat (2 x 2)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Bytes** | 61624 | 78856 | 225240 | 1681728 | 3349448 |

**Encryption Memory Stat (3 x 3)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Bytes** | 61624 | 78856 | 225240 | 1681728 | 3349448 |

**Encryption Memory Stat (4 x 4)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Bytes** | 61624 | 78856 | 225240 | 1681728 | 3349448 |

## Encryption Memory Stat (5 x 5)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 65256 | 69936 | 136536 | 851136 | 7964112 | 8730224 |

## Encryption Memory Stat (6 x 6)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 65240 | 69848 | 127416 | 774152 | 6780992 | 9278024 |

## Encryption Memory Stat (7 x 7)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 65256 | 69848 | 128040 | 757704 | 6419600 | 11842448 |

## Encryption Memory Stat (8 x 8)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 65216 | 69848 | 127416 | 719560 | 6040552 | 15096856 |

## Encryption Memory Stat (9 x 9)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 65192 | 69848 | 127416 | 711336 | 5827880 | 17119376 |

## Encryption Memory Stat (10 x 10)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 65232 | 69848 | 119192 | 703200 | 5693984 | 26489856 |

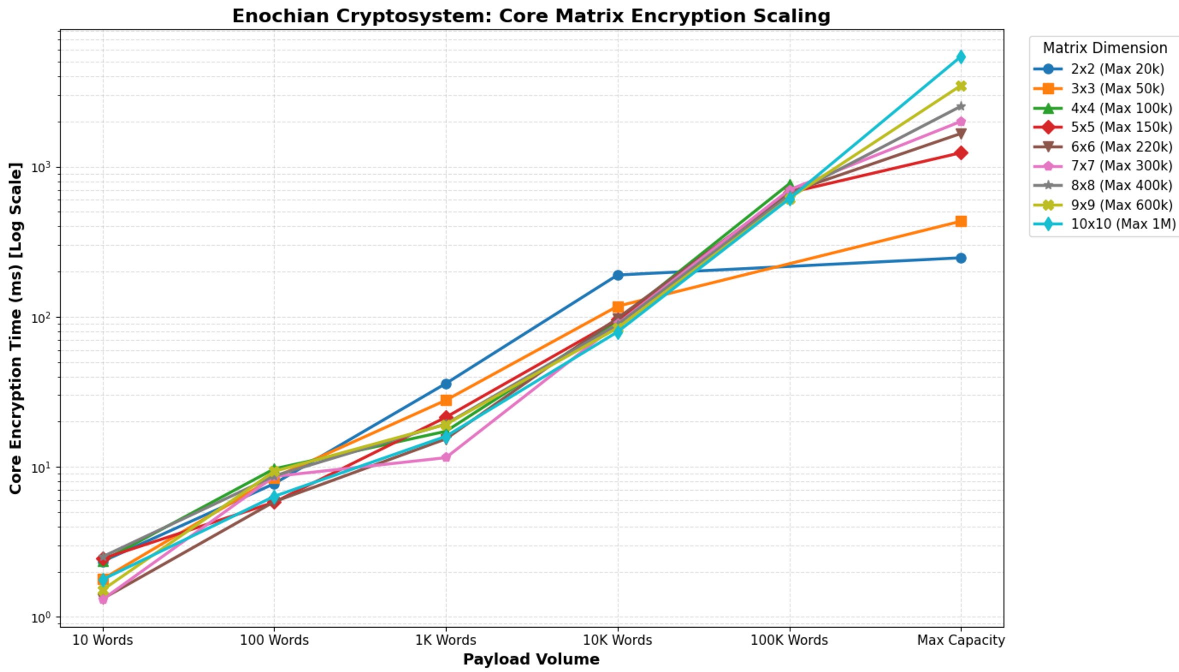


**Figure 9 – Core Matrix Encryption Scaling Benchmark**

The above logarithmic line chart maps the isolated core encryption latencies as payload volume scale from 10 words to maximum architectural capacity. By consolidating all matrix dimensions into a single visualization, a critical performance characteristic of the framework is exposed: large-dimension matrices become highly computationally efficient at massive payload scales. For instance, while the 5 x 5 matrix requires 670.5 ms to encrypt a 100,000-word payload, the 10 x 10 matrix encrypts the exact same volume in only 610.6 ms. This verifies that despite the polynomial complexity inherently associated with initializing larger continuous arrays, their volumetric block capacity significantly reduces the per-word encryption latency at scale, rendering higher-dimension matrices structurally superior for massive data transmissions.

### 6) Encryption CPU Stats

The execution latency provides a temporal measure of algorithmic speed but it is equally essential to evaluate the localized processor load to ensure the cryptosystem does not induce hardware throttling or resource exhaustion. This spreadsheet analyzes the raw CPU processing statistics generated during the encryption phase across all payload volumes and matrix dimensions.

## Encryption CPU Stat (2 x 2)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 20000 Words (Maximum) |
|---|---|---|---|---|---|
| **Time (ms)** | 4343.75 | 1312.5 | 1140.625 | 1468.75 | 1750 |

## Encryption CPU Stat (3 x 3)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| **Time (ms)** | 1250 | 1234.375 | 1265.625 | 1328.125 | 2078.125 |

## Encryption CPU Stat (4 x 4)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Time (ms)** | 1421.875 | 968.75 | 1312.5 | 1515.625 | 1718.75 |

## Encryption CPU Stat (5 x 5)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 1437.5 | 1296.875 | 1218.75 | 1484.375 | 1875 | 1984.375 |

## Encryption CPU Stat (6 x 6)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 1328.125 | 1546.875 | 1484.375 | 1234.375 | 2093.75 | 3031.25 |

## Encryption CPU Stat (7 x 7)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 1390.625 | 1125 | 1218.75 | 1328.125 | 2281.25 | 2562.5 |

## Encryption CPU Stat (8 x 8)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 1296.875 | 1390.625 | 1515.625 | 1515.625 | 1875 | 3796.875 |

## Encryption CPU Stat (9 x 9)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 1406.25 | 1203.125 | 1234.375 | 1359.375 | 1921.875 | 4281.25 |

## Encryption CPU Stat (10 x 10)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 1093.75 | 1000 | 1062.5 | 1015.625 | 1781.25 | 5406.25 |

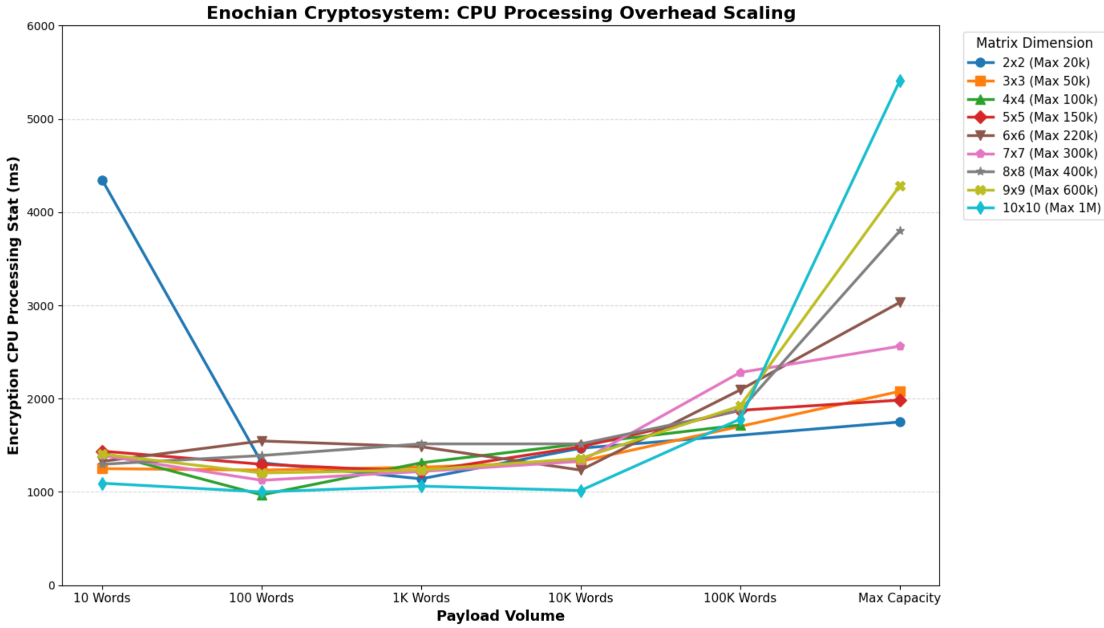


**Figure 9 – CPU Processing Overhead Scaling**

The unified line chart displays the processor usage footprint of the Enochian Cryptosystem. The data renders a highly stabilized and efficient baseline: for standard payload volumes ranging from 100 to 1,000 words, processor overhead remains remarkable flat across all matrix configurations, consistently operating between 1,000 and 1,500 processing time in millisecond. A distinct cold-start initialization anomaly is observed in the 2 x 2 matrix for the 10-word payload, which is indicative of the initial thread allocation overhead bypassing the extremely short mathematical execution time.

Essentially, the chart demonstrates predictable, linear scaling as the matrices are pushed to their maximum volumetric limits. Even at the absolute extreme, encrypting 1,000,000 words via a 10 x 10 continuous array, the CPU stat peaks at an entirely manageable 5,406.25 ms. This mathematically validates that the continuous matrix architecture defers significant processor load exclusively to massive payload scenarios, maintaining a highly lightweight execution profile for standard, real-time secure transmission.

### 7) Decryption Stats

The decryption lifecycle of the Enochian Cryptosystem requires reversing the initial non-linear transformations applied during encryption. Analyzing the execution metrics of this recovery process is critical for understanding the systemic asymmetry of the framework. The dataset isolates ten distinct computational phases, from initial Package Delivery and Key Encapsulation through to Reverse Shifts and Finalization, across all matrix dimensions and scaling payload volumes.

## 2 x 2 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 20000 Words (Maximum) |
|---|---|---|---|---|---|
| Step 1: Package Delivery | 91.6899 | 112.4079 | 4249.2426 | 2776.2637 | 4070.9842 |
| Step 2: Signature Verification | 1035.2605 | 1096.6944 | 1113.8332 | 1039.985 | 899.4879 |
| Step 3: Key Decapsulation | 864.8617 | 870.4321 | 1108.594 | 828.0006 | 976.1269 |
| Step 4a: Sorting | 0.1101 | 0.1647 | 27.8133 | 28.2099 | 27.5692 |
| Step 4b: Binary Search | 941.2232 | 995.3483 | 5828.913 | 1072.9686 | 1168.4889 |
| Step 5: Regeneration | 25.9605 | 8.6446 | 204.9384 | 23.6216 | 3.6916 |
| Step 6: Core Decryption | 12.8079 | 10.4481 | 163.9146 | 188.9483 | 258.2487 |
| Step 7: Reverse Shifts | 7.0562 | 45.0892 | 464.4449 | 8917.7033 | 13171.6652 |
| Step 8: Finalization | 9.7328 | 8.4774 | 58.487 | 26.609 | 22.9146 |
| Total Time (ms) | 2988.6938 | 3147.7067 | 13220.181 | 14902.31 | 20599.1772 |

## 3 x 3 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| Step 1: Package Delivery | 54.5166 | 63.2934 | 213.5663 | 1578.7848 | 7676.7158 |
| Step 2: Signature Verification | 830.8479 | 748.3686 | 1051.6025 | 837.7321 | 1769.1801 |
| Step 3: Key Decapsulation | 1095.3744 | 750.2778 | 783.7961 | 1127.3795 | 987.0138 |
| Step 4a: Sorting | 0.0756 | 0.1304 | 0.3515 | 2.1862 | 29.3619 |
| Step 4b: Binary Search | 1061.0863 | 911.399 | 1094.2688 | 1136.2587 | 1041.1991 |
| Step 5: Regeneration | 6.1244 | 9.2083 | 8.3232 | 5.1984 | 4.5967 |
| Step 6: Core Decryption | 13.0067 | 11.8312 | 20.6439 | 135.4037 | 489.2268 |
| Step 7: Reverse Shifts | 3.2262 | 31.7765 | 430.4642 | 9929.9359 | 52602.0343 |
| Step 8: Finalization | 5.3201 | 5.1536 | 6.0882 | 22.147 | 40.22 |
| Total Time (ms) | 3069.5782 | 2531.4388 | 3609.1047 | 14774.9893 | 64639.5485 |

## 4 x 4 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| Step 1: Package Delivery | 125.8262 | 63.9089 | 185.907 | 1446.5513 | 15998.3073 |
| Step 2: Signature Verification | 2539.268 | 665.3159 | 998.7974 | 973.6479 | 723.9862 |
| Step 3: Key Decapsulation | 1359.1397 | 1848.8965 | 882.5242 | 990.4589 | 1209.8635 |
| Step 4a: Sorting | 0.1034 | 0.1172 | 0.2888 | 0.9826 | 31.8815 |
| Step 4b: Binary Search | 1430.4724 | 1375.1287 | 1042.8637 | 1421.8159 | 2664.6253 |
| Step 5: Regeneration | 20.8736 | 18.8037 | 15.6997 | 17.0229 | 16.0177 |
| Step 6: Core Decryption | 15.2234 | 16.518 | 22.0061 | 122.1066 | 784.3912 |
| Step 7: Reverse Shifts | 3.3773 | 41.2199 | 392.2296 | 10090.0034 | 157744.1332 |
| Step 8: Finalization | 6.8331 | 6.7749 | 8.3033 | 26.8374 | 123.9252 |
| Total Time (ms) | 5501.1171 | 4036.6837 | 3548.6198 | 15089.4269 | 179297.1311 |

## 5 x 5 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Step 1: Package Delivery | 83.0261 | 60.8018 | 153.2374 | 1312.9379 | 13254.9251 | 21136.285 |
| Step 2: Signature Verification | 972.4284 | 961.1714 | 2097.5262 | 1096.9075 | 864.4591 | 938.9756 |
| Step 3: Key Decapsulation | 1171.034 | 1238.6735 | 1116.2732 | 890.8829 | 906.7947 | 1042.0847 |
| Step 4a: Sorting | 0.066 | 0.0869 | 0.2155 | 0.7257 | 30.1548 | 30.5935 |
| Step 4b: Binary Search | 887.4685 | 850.1216 | 1058.5366 | 1091.9829 | 1104.4168 | 1535.767 |
| Step 5: Regeneration | 19.7466 | 17.184 | 17.6199 | 17.7303 | 18.0629 | 17.4858 |
| Step 6: Core Decryption | 23.6567 | 34.8284 | 32.8761 | 126.5497 | 718.9079 | 1145.0183 |
| Step 7: Reverse Shifts | 5.3901 | 37.183 | 516.0528 | 10169.562 | 163189.3838 | 336911.0135 |
| Step 8: Finalization | 9.8553 | 8.2764 | 5.8182 | 25.5726 | 62.5434 | 61.3265 |
| Total Time (ms) | 3172.6717 | 3208.327 | 4998.1559 | 14732.85115 | 180149.6485 | 362818.5499 |

## 6 x 6 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Step 1: Package Delivery | 223.3 | 86.3704 | 149.082 | 1220.89 | 12892.0245 | 23457.629 |
| Step 2: Signature Verification | 1067.3322 | 823.9128 | 1073.2586 | 79.0745 | 805.7586 | 1123.1038 |
| Step 3: Key Decapsulation | 936.0551 | 967.2849 | 863.2278 | 1173.64 | 837.6831 | 2703.2924 |
| Step 4a: Sorting | 0.0918 | 0.0565 | 0.1564 | 0.7392 | 25.1568 | 31.3418 |
| Step 4b: Binary Search | 1100.1368 | 1072.08 | 985.5792 | 1079.3968 | 1180.0574 | 1373.224 |
| Step 5: Regeneration | 22.0917 | 30.732 | 20.7509 | 21.7328 | 21.6698 | 17.6443 |
| Step 6: Core Decryption | 34.3628 | 46.7143 | 42.2954 | 148.2905 | 756.9556 | 1753.1013 |
| Step 7: Reverse Shifts | 4.3584 | 40.8843 | 401.0485 | 9768.9228 | 160848.5864 | 546103.6699 |
| Step 8: Finalization | 11.7796 | 5.3517 | 7.1092 | 22.0039 | 68.8891 | 209.5156 |
| Total Time (ms) | 3399.5084 | 3073.3869 | 3542.508 | 14234.6905 | 177436.7813 | 576772.5221 |

## 7 x 7 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Step 1: Package Delivery | 213.6364 | 82.1494 | 160.4221 | 1275.8009 | 12235.5435 | 39110.2343 |
| Step 2: Signature Verification | 888.1336 | 946.2439 | 1715.7974 | 1260.5934 | 751.8433 | 712.9354 |
| Step 3: Key Decapsulation | 1001.0216 | 1386.0285 | 1030.2593 | 1502.1675 | 2602.1786 | 887.0552 |
| Step 4a: Sorting | 0.0773 | 0.0897 | 0.1652 | 0.5806 | 25.1959 | 40.3363 |
| Step 4b: Binary Search | 868.363 | 1020.0604 | 972.8519 | 1050.8049 | 1138.9933 | 892.9039 |
| Step 5: Regeneration | 40.2472 | 30.5956 | 27.6 | 34.4611 | 22.8234 | 42.3586 |
| Step 6: Core Decryption | 60.1165 | 36.0091 | 56.8652 | 249.8805 | 748.9984 | 2583.4989 |
| Step 7: Reverse Shifts | 3.7657 | 43.1469 | 431.4514 | 10382.8284 | 165968.2815 | 1271904.754 |
| Step 8: Finalization | 25.9649 | 13.7805 | 5.9791 | 44.991 | 59.6998 | 152.3537 |
| Total Time (ms) | 3101.3262 | 3558.104 | 4401.3916 | 15802.1083 | 183553.5577 | 1316326.43 |

## 8 x 8 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Step 1: Package Delivery | 67.8126 | 60.9919 | 192.4145 | 1523.3985 | 12664.5378 | 52137.237 |
| Step 2: Signature Verification | 619.2203 | 669.9222 | 781.7179 | 866.376 | 983.7826 | 830.4484 |
| Step 3: Key Decapsulation | 944.1924 | 1003.862 | 889.6778 | 981.57 | 916.2464 | 1553.1889 |
| Step 4a: Sorting | 0.0721 | 0.0531 | 0.143 | 0.37 | 2.7033 | 39.5234 |
| Step 4b: Binary Search | 909.2275 | 784.5725 | 719.3866 | 986.8005 | 915.2778 | 1769.1406 |
| Step 5: Regeneration | 29.2425 | 33.4255 | 30.7578 | 29.6125 | 31.4489 | 64.275 |
| Step 6: Core Decryption | 63.6125 | 60.8241 | 62.0702 | 155.185 | 676.5822 | 3052.2794 |
| Step 7: Reverse Shifts | 4.7341 | 37.9635 | 428.3775 | 9816.125 | 162642.06 | 1986747.382 |
| Step 8: Finalization | 6.985 | 5.568 | 6.0323 | 31.4773 | 45.2154 | 170.5063 |
| Total Time (ms) | 2645.0991 | 2657.107 | 3110.5776 | 14390.9148 | 178877.8544 | 2046363.981 |

## 9 x 9 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Step 1: Package Delivery | 6610.2331 | 66.4639 | 168.028 | 1271.5795 | 11760.3003 | 182054.8934 |
| Step 2: Signature Verification | 839.4659 | 1608.3174 | 1167.7948 | 829.3732 | 905.6036 | 1308.7997 |
| Step 3: Key Decapsulation | 1039.5719 | 707.4735 | 751.7113 | 1004.2239 | 1073.7189 | 953.205 |
| Step 4a: Sorting | 0.0739 | 0.0789 | 0.1323 | 0.4031 | 1.9762 | 64.6632 |
| Step 4b: Binary Search | 1132.503 | 1204.1126 | 896.5269 | 993.682 | 1366.715 | 2037.0292 |
| Step 5: Regeneration | 185.8333 | 39.4789 | 33.8177 | 36.2022 | 25.7161 | 60.9575 |
| Step 6: Core Decryption | 148.6123 | 58.3184 | 65.6908 | 151.0575 | 798.4424 | 5570.9958 |
| Step 7: Reverse Shifts | 62.6528 | 45.7403 | 424.6473 | 10213.5739 | 167359.1445 | 4354328.81 |
| Step 8: Finalization | 68.6097 | 5.8278 | 5.6851 | 46.49 | 46.444 | 1623.6167 |
| Total Time (ms) | 10087.556 | 3735.8117 | 3514.0342 | 14546.5853 | 183338.061 | 4548002.971 |

## 10 x 10 Matrix Decryption

| Cryptographic Phase | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| Step 1: Package Delivery | 189.9702 | 81.9879 | 685.2124 | 1314.292 | 10383.0253 | 402750.434 |
| Step 2: Signature Verification | 698.0867 | 804.9172 | 709.938 | 979.3125 | 1172.5627 | 5120.2758 |
| Step 3: Key Decapsulation | 1086.7891 | 961.5851 | 1174.7664 | 919.1437 | 7958.1076 | 1332.4676 |
| Step 4a: Sorting | 0.0739 | 0.0553 | 0.0939 | 0.2998 | 2.0487 | 169.5541 |
| Step 4b: Binary Search | 1084.6437 | 879.8308 | 1180.6599 | 920.5106 | 1162.1417 | 1367.58 |
| Step 5: Regeneration | 32.8597 | 34.5109 | 32.3501 | 30.4898 | 30.3336 | 715.5672 |
| Step 6: Core Decryption | 53.4812 | 59.403 | 62.2567 | 141.4582 | 642.2142 | 8529.141 |
| Step 7: Reverse Shifts | 7.3326 | 32.9737 | 632.0802 | 8053.2444 | 125349.5033 | 9986342.033 |
| Step 8: Finalization | 4.4605 | 4.1643 | 30.4694 | 26.9462 | 68.7272 | 1164.8189 |
| Total Time (ms) | 3157.6976 | 2859.4282 | 4507.827 | 12385.6792 | 146768.6643 | 10407491.87 |

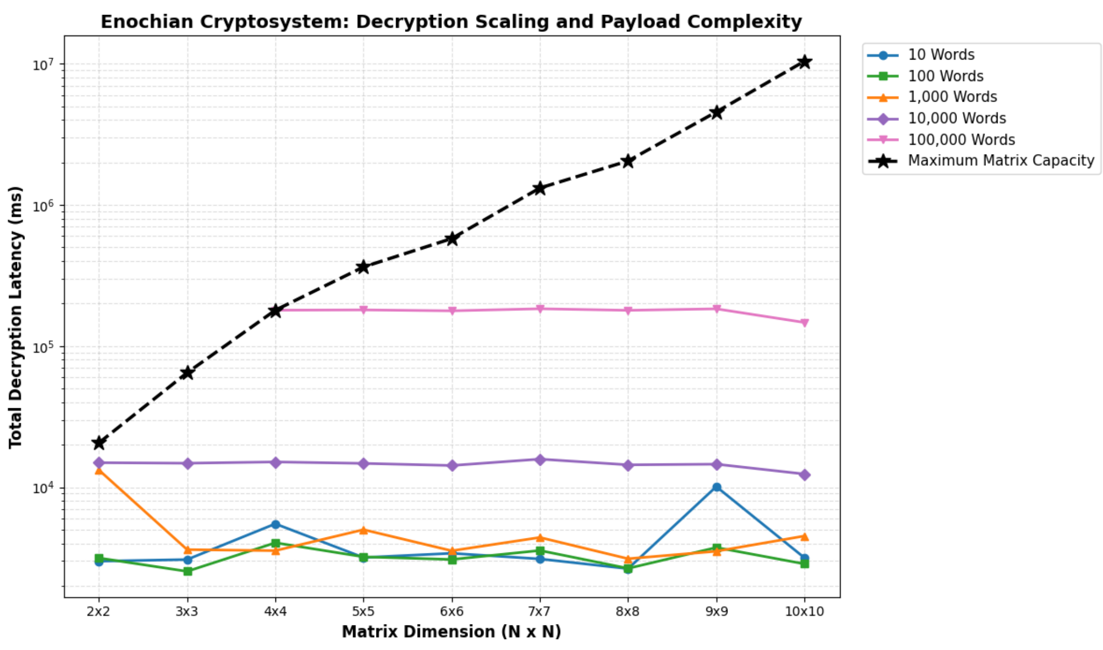


**Figure 10 – Decryption Scaling and Payload Complexity**

This logarithmic link chart maps the total execution time of the complete decryption cycle for evaluating the absolute temporal boundaries of the framework. For constant, moderate payloads up to 10,000 words, the total decryption time remains relatively static across the geometric growth of the matrices. However, plotting the absolute maximum capacity limits maps the definitive $O(N^3)$ polynomial scaling wall. Since payload capacities peak 100,000 words in the

5 x 5 configuration and scale up to 1,000,000 words in the 10 x 10 boundary, the decryption latency curves aggressively upward, requiring approximately 10,407 seconds for total plaintext recovery at absolute maximum capacity.

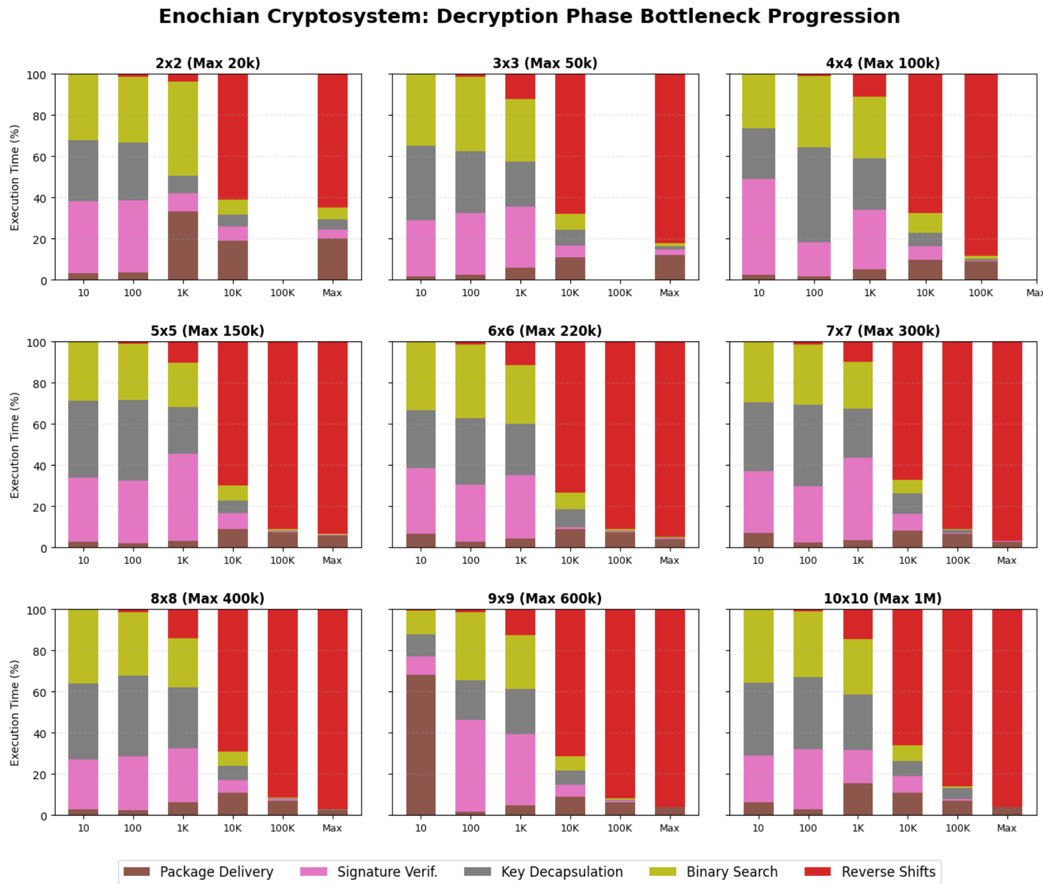


**Figure 11 – Decryption Phase Bottleneck Progression for all Matrices**

The 3 x 3 grid normalizes the total decryption latency into a 100% stacked percentage distribution, isolating the proportional CPU allocation for the five heaviest computational phases. The visualization mathematically proves a distinct operational asymmetry compared to the encryption phase. At smaller payload volumes, overheads such as Signature Verification, Key Decapsulation, and Binary Search consume the majority of the processing cycle.

Notably, the dataset captured an observable I/O anomaly within the 9 x 9 matrix at 10 words, where "Package Delivery" briefly overwhelmed the local mathematics due to localized transmission latency. However, as payloads scale toward the maximum theoretical capacity of each matrix, "Step 7: Reverse Shifts" exponentially overwhelms the execution cycle. In the 10 x

10 matrix operating at its 1,000,000-word limit, this single un-mapping operation accounts for over 95% of the processor's time, establishing the primary decryption bottleneck.

**8) Decryption Speed Benchmark**

The absolute decryption throughput and relative execution speed of the Enochian Cryptosystem during the ciphertext unmapping phase is evaluated in this part. Since the cryptosystem uses a continuous matrix architecture, the computational effort required to reverse the non-linear transformation scales differently than traditional prime-based or lattice-based systems. Its decryption throughput is mapped across exponentially scaling payload volumes against RSA-2048, ECC Curve25519, and NIST post-quantum standard ML-KEM to evaluate the algorithm's standing.

**2 x 2 Matrix Decryption Speed**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Enochian** | 858.84 | 8231.16 | 5008.71 | 44070.26 | 64577.29 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 386.6471 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 340485.8272 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 185169.05 |
| **Decryption Speed Ratio (Times)** | | | | | |
| **RSA** | 7.95 | 31.66 | 12.79 | 118.38 | 167.02 |
| **ECC** | 2.62 | 1.16 | 0.072 | 0.081 | 0.19 |
| **Kyber** | 247.5 | 19.88 | 0.961 | 0.913 | 0.349 |

## 3 x 3 Matrix Decryption Speed

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| **Enochian** | 615.07 | 5155.86 | 27756.38 | 43004.73 | 59589.95 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 359.6002 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 51475.1064 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 210927.92 |
| **Decryption Speed Ratio (Times)** | | | | | |
| **RSA** | 5.69 | 19.83 | 70.87 | 115.52 | 165.71 |
| **ECC** | 1.88 | 0.729 | 0.4 | 0.079 | 1.158 |
| **Kyber** | 177.25 | 12.45 | 5.33 | 0.891 | 0.282 |

## 4 x 4 Matrix Decryption Speed

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Enochian** | 459.82 | 3208.62 | 22130.23 | 40489.21 | 63153.44 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 |
| **Decryption Speed Ratio (Times)** | | | | | |
| **RSA** | 4.26 | 12.34 | 56.5 | 108.76 | 159.07 |
| **ECC** | 1.41 | 0.453 | 0.319 | 0.075 | 0.17 |
| **Kyber** | 132.51 | 7.75 | 4.25 | 0.839 | 0.119 |

## 5 x 5 Matrix Decryption Speed

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 338.17 | 1378.26 | 13626.92 | 35835.72 | 63247.32 | 59577.21 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 | 392.388 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 | 563631.8826 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 | 583049.64 |
| **Decryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 3.13 | 5.3 | 34.79 | 96.26 | 159.3 | 151.83 |
| **ECC** | 1.03 | 0.195 | 0.197 | 0.066 | 0.171 | 0.106 |
| **Kyber** | 97.46 | 3.33 | 2.62 | 0.742 | 0.119 | 0.102 |

## 6 x 6 Matrix Decryption Speed

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 203.71 | 1006.12 | 10072.02 | 29111.78 | 57131.3 | 54337.99 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 | 404.5598 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 | 561753.2713 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 | 563693.53 |
| **Decryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 1.89 | 3.87 | 25.72 | 78.2 | 143.9 | 134.31 |
| **ECC** | 0.623 | 0.142 | 0.145 | 0.054 | 0.154 | 0.097 |
| **Kyber** | 58.71 | 2.43 | 1.93 | 0.603 | 0.107 | 0.096 |

## 7 x 7 Matrix Decryption Speed

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **300000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Enochian** | 133.07 | 1305.23 | 7245.2 | 16740 | 55980.09 | 48766.81 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 | 344.5208 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 | 85419.1742 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 | 429187.95 |
| **Decryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 1.23 | 5.02 | 18.5 | 44.97 | 141 | 141.55 |
| **ECC** | 0.407 | 0.184 | 0.104 | 0.031 | 0.151 | 0.571 |
| **Kyber** | 38.35 | 3.15 | 1.39 | 0.347 | 0.105 | 0.114 |

## 8 x 8 Matrix Decryption Speed

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **400000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Enochian** | 94.32 | 739.48 | 6540.98 | 26400.75 | 60694.77 | 53923.31 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 | 405.7197 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 | 1037030.793 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 | 743436.53 |
| **Decryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 0.873 | 2.84 | 16.7 | 70.92 | 152.87 | 132.91 |
| **ECC** | 0.288 | 0.104 | 0.094 | 0.049 | 0.164 | 0.052 |
| **Kyber** | 27.18 | 1.79 | 1.26 | 0.547 | 0.114 | 0.073 |

## 9 x 9 Matrix Decryption Speed

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 47.1 | 754.48 | 6089.13 | 26751.4 | 50687.44 | 43690.93 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 | 403.9685 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 | 611810.9279 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 | 348749.37 |
| **Decryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 0.436 | 2.9 | 15.55 | 71.86 | 127.67 | 108.15 |
| **ECC** | 0.144 | 0.107 | 0.088 | 0.049 | 0.137 | 0.071 |
| **Kyber** | 13.57 | 1.82 | 1.17 | 0.554 | 0.095 | 0.125 |

## 10 x 10 Matrix Decryption Speed

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Enochian** | 149.59 | 757.54 | 6328.64 | 28283.97 | 62381.06 | 47108.5 |
| **RSA-2048** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 397.0285 | 406.387 |
| **ECC/X25519** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 370916.748 | 798969.8258 |
| **Kyber** | 3.47 | 414.04 | 5209.3 | 48268.95 | 532830.67 | 533697.96 |
| **Decryption Speed Ratio (Times)** | | | | | | |
| **RSA** | 1.38 | 2.91 | 16.16 | 75.97 | 157.12 | 115.92 |
| **ECC** | 0.457 | 0.107 | 0.091 | 0.052 | 0.168 | 0.059 |
| **Kyber** | 43.11 | 1.83 | 1.21 | 0.586 | 0.117 | 0.088 |

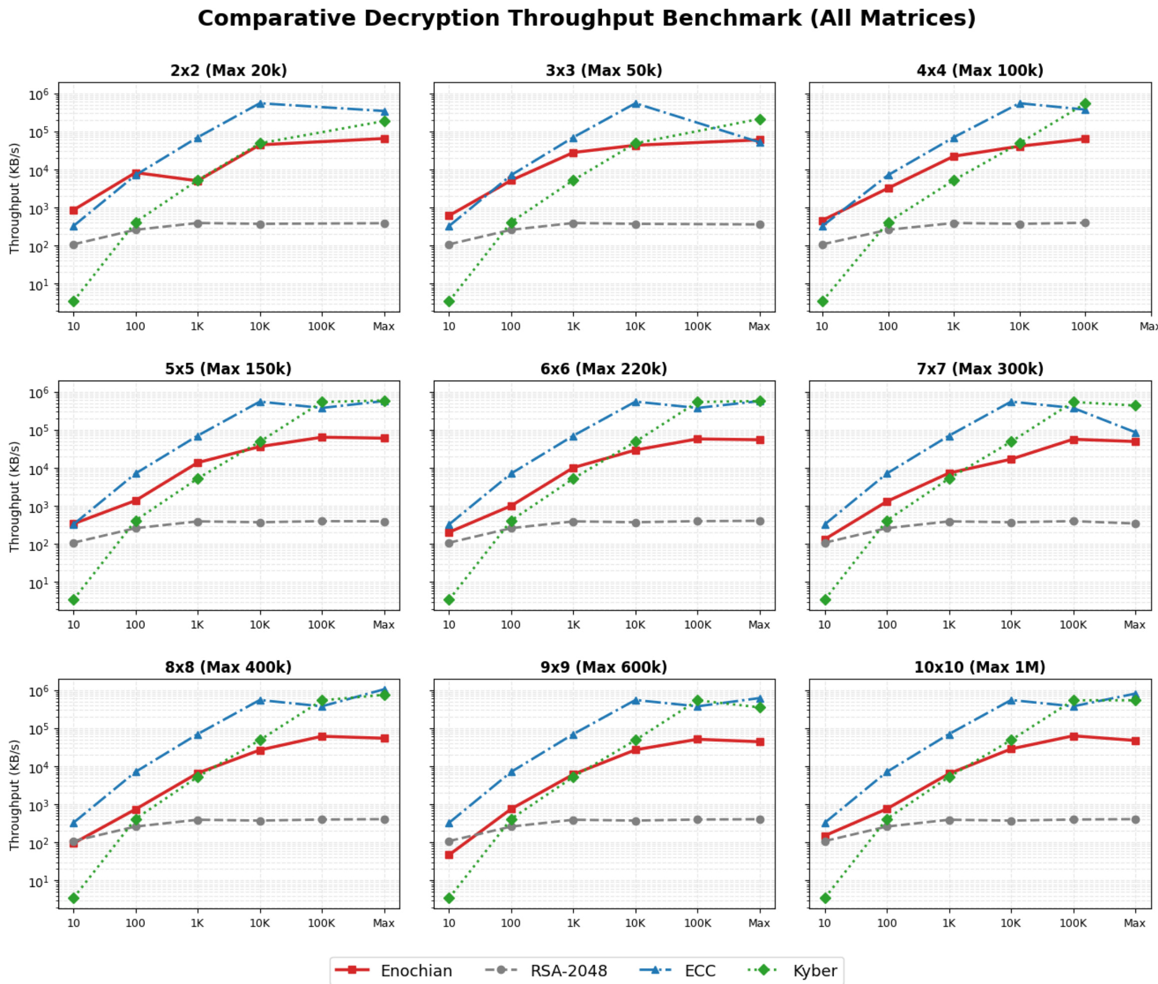


**Figure 12 – Comparative Decryption Throughput Benchmark**

The faceted 3 x 3 grid renders the absolute decryption throughput of the framework across all nine geometric configurations. The visualization demonstrates that Enochian consistently surpasses the throughput limits of legacy RSA-2048 operations by using a logarithmic Y-axis to normalize the performance disparity. While highly optimized, lightweight protocols, ECC and Kyber, mathematically dominate the absolute throughput ceiling which is frequently operating in the hundreds of thousands of KB/s at large scales, the Enochian continuous array establishes a robust middle-tier performance envelope. At maximum capabilities, the framework reliably processes between 40,000 and 60,000 KB/s, establishing it as a highly viable, stable decryption mechanism for continuous data streams.

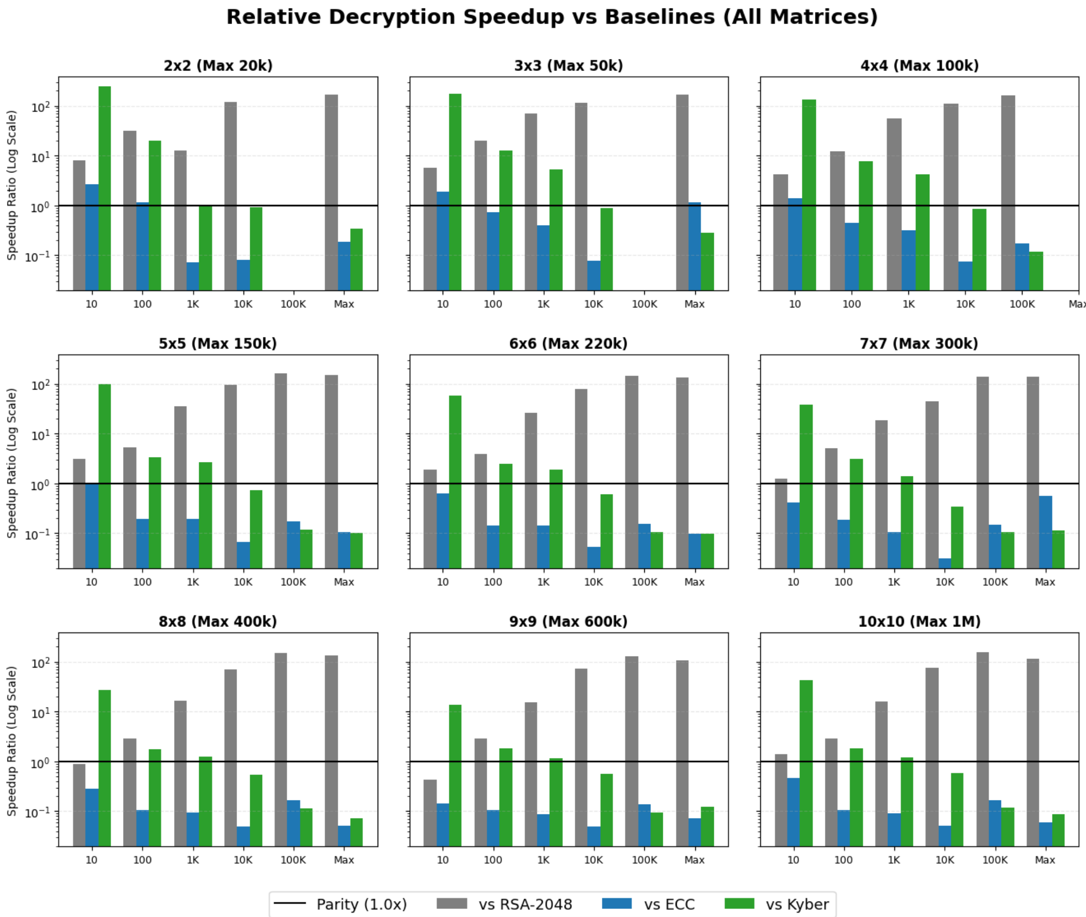


**Figure 13 – Relative Decryption Speedup vs. Baselines**

The faceted diverging graphs translates the raw throughput metrics into a relative speedup multiplier, using a parity line at 1.0x to clearly define the boundaries of computational superiority. The data unequivocally proves that the Enochian decryption process is vastly superior to RSA-2048, achieving speedup multipliers as high as 167x at peak efficiency. Conversely, points plotting below the parity baseline properly map the architectural domains where the Enochian system produces to modern lattice and elliptic curve efficiencies, demonstrating exactly how the required "Reverse Shift" bottleneck influences the algorithm's standing against strictly optimized asymmetric standards.

### 9) Decryption Security Entropy

The maximum ciphertext entropy ensures the indistinguishability and resistance to statistical cryptanalysis but the lossless reconstruction of the original plaintext upon decryption is required to be guaranteed for a robust cryptosystem. The output plaintext was subjected to Shannon Entropy analysis for verifying the semantic integrity of the Enochian algorithm's decryption phase mathematically. Standard natural English text possesses a well-established entropy boundary of approximately 3.5 to 4.2 bits per character, dictated by linguistic structural rules and repeating character frequencies. Successfully returning the data to this specific mathematical domain proves that the decryption matrix array un-maps the ciphertext without inducing data corruption or loss.

## 2 x 2 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 20000 Words (Maximum) |
|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.9039 | 3.6427 | 4.0436 | 3.9317 | 3.767 |

## 3 x 3 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.8983 | 3.8557 | 3.5977 | 3.7909 | 3.7404 |

## 4 x 4 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.5815 | 3.7277 | 4.0441 | 3.5983 | 3.7909 |

## 5 x 5 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.7731 | 4.0172 | 3.7901 | 3.7404 | 3.7404 | 4.0802 |

## 6 x 6 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.5338 | 3.9554 | 3.957 | 3.9337 | 3.767 | 3.5982 |

## 7 x 7 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.0041 | 4.0459 | 3.965 | 3.6825 | 3.8336 | 3.9333 |

## 8 x 8 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.9284 | 3.8969 | 3.8922 | 3.7908 | 3.6812 | 3.5982 |

## 9 x 9 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.0949 | 3.7555 | 3.7316 | 3.7411 | 3.8503 | 3.8347 |

## 10 x 10 Decryption Security

**(Target Entropy for English ~ 3.5 to 4.2)**

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Decrypted Entropy** | 3.1609 | 3.7473 | 3.6015 | 3.8358 | 4.0171 | 3.9154 |

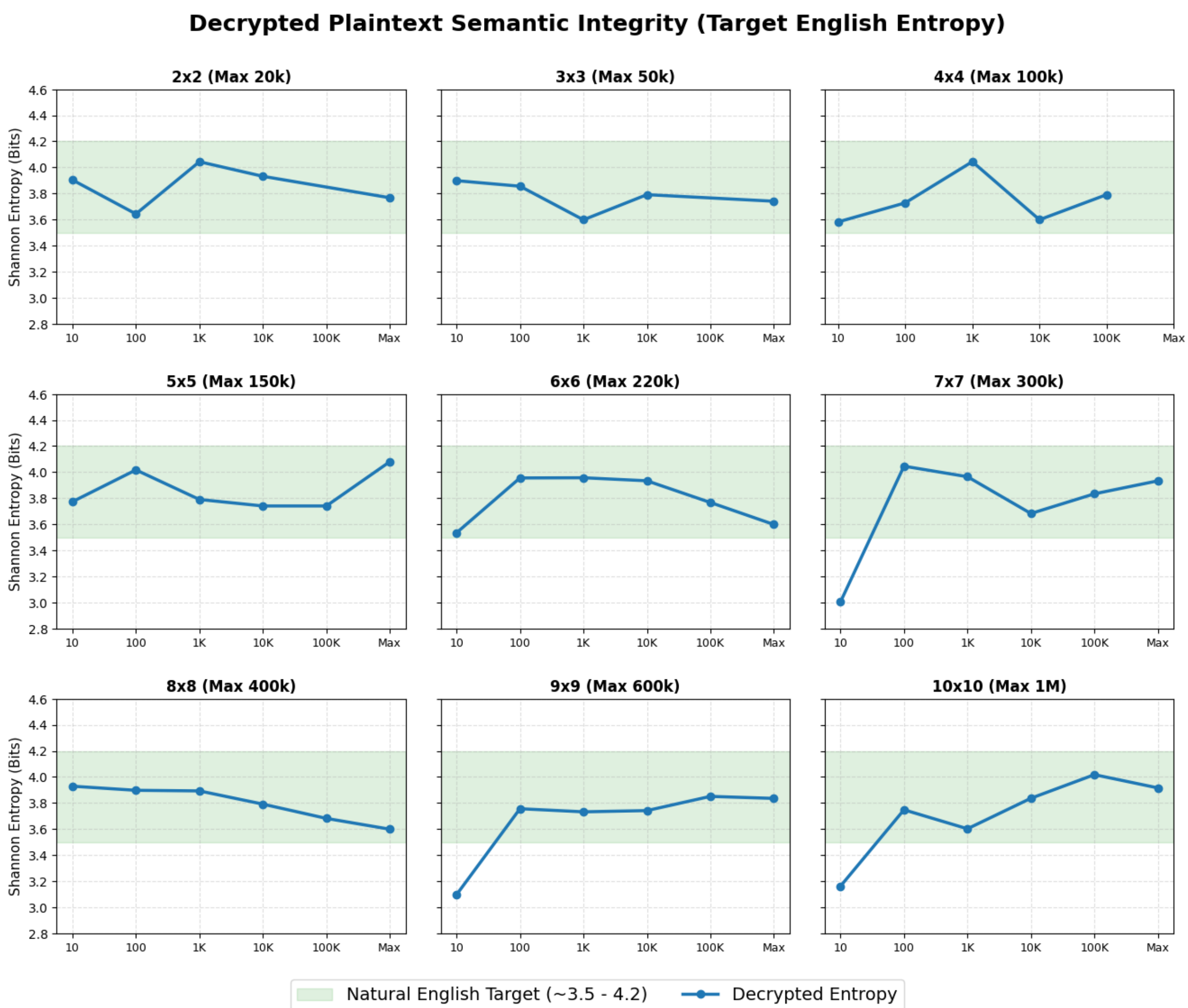


**Figure 14 – Decrypted Plaintext Semantic Integrity**

The faceted grids render the measured Shannon Entropy of the decrypted plaintext across all matrix dimensions and payload sizes. The shaded green region represents the optimal 3.5 to 4.2 natural English target band. The visualization provides absolute confirmation of lossless recovery: the recovery entropy reliably stabilizes within the target boundaries across all matrices once statistical significance is reached. Minor deviates falling slightly below the 3.5 bits threshold which is specifically observed at the 10-word payload in the 7 x 7, 9 x 9, and 10 x 10 matrices, are natural statistical anomalies because a 10-word sample lacks the requisite volumetric length to express a complete natural language frequency distribution. The entropy trajectories flatten directly into the linguistic center when the payload volumes expand to 100 words and beyond, and it proves that the framework executes a flawless decryption lifecycle structurally.

### 10) Decryption Quantum Safety

The post-quantum security of a cryptographic framework is a systemic property defined by the complexity of its key space. During the decryption phase, the primary threat model assumes an attacker has intercepted the ciphertext and is using quantum mechanisms to reverse-engineer the continuous matrix array. Since the Enochian Cryptosystem' core operations do not rely on integer factorization or the discrete logarithm problem, the ciphertext is inherently immune to Shor's algorithm. The only viable quantum vector is Grover's algorithm, which applies a quadratic speedup to brute-force key space searches, effectively halving the classical bit security of the matrix.

**Post-Quantum Resistance Proof**

| **Matrix Size: (Field 21)** | 2x2 | 3x3 | 4x4 | 5x5 | 6x6 | 7x7 | 8x8 | 9x9 | 10x10 |
|---|---|---|---|---|---|---|---|---|---|
| **Classical Key Strength** | 18 | 40 | 70 | 110 | 158 | 215 | 281 | 356 | 439 |
| **Shor's Attack** | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant | Resistant |
| **Grover's Attack** | 9 | 20 | 35 | 55 | 79 | 108 | 141 | 178 | 220 |
| **Required Standard** | 128 | 128 | 128 | 128 | 128 | 128 | 128 | 128 | 128 |
| **Result** | Weak | Weak | Weak | Weak | Weak | Weak | Secure | Secure | Secure |

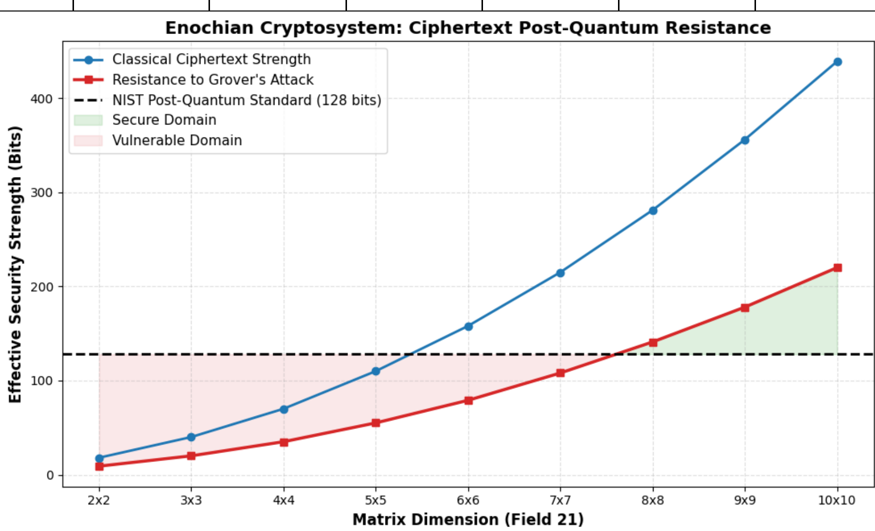


**Figure 15 – Ciphertext Post-Quantum Security Thresholds**

The graph maps the theoretical resilience of the Enochian ciphertext against Grover's quantum search algorithm. The visualization demonstrates that the effective quantum resistance mirrors the encryption phase, confirming that the key space constraints remain absolute regardless of the direction of the cryptographic operation. Matrices sized 2 x 2 through 7 x 7 fail to meet the required NIST 128-bit post-quantum standard, rendering their ciphertexts vulnerable to advanced quantum brute-forcing. The visual intersection explicitly confirms that an 8 x 8 configuration acts as the absolute baseline for secure ciphertext transmission, while the 10 x 10 framework provides a highly secure ceiling of 220 effective bits, fully insulating the decrypted data from next-generation adversarial models.

**11) Decryption Time Stats**

The overall decryption latency is heavily influenced by non-linear un-mapping operations, but it is critical to isolate the core mathematical decryption phase to evaluate the framework's volumetric substitution efficiency. The decryption time stats isolate the execution time in milliseconds required strictly for the core matrix decryption algorithm, eliminating the peripheral systemic overheads to reveal the raw computational performance of the continuous array.

**Decryption Time Stat (2 x 2)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Time (ms)** | 12.8079 | 10.4481 | 163.9146 | 188.9483 | 258.2487 |

**Decryption Time Stat (3 x 3)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **50000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Time (ms)** | 13.0067 | 11.8312 | 20.6439 | 135.4037 | 489.2268 |

**Decryption Time Stat (4 x 4)**

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Time (ms)** | 15.2234 | 16.518 | 22.0061 | 122.1066 | 784.3912 |

## Decryption Time Stat (5 x 5)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 23.6567 | 34.8264 | 32.8761 | 126.5497 | 718.9079 | 1145.0183 |

## Decryption Time Stat (6 x 6)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 34.3628 | 46.7143 | 42.2954 | 148.2905 | 756.9556 | 1753.1013 |

## Decryption Time Stat (7 x 7)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 60.1165 | 36.0091 | 56.8652 | 249.8805 | 748.9984 | 2583.4989 |

## Decryption Time Stat (8 x 8)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 63.6126 | 60.8241 | 62.0702 | 155.185 | 676.5822 | 3052.2794 |

## Decryption Time Stat (9 x 9)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 148.6123 | 58.3184 | 65.6908 | 151.0575 | 798.4424 | 5570.9958 |

## Decryption Time Stat (10 x 10)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 53.4812 | 59.403 | 62.2567 | 141.4582 | 642.2142 | 8529.141 |

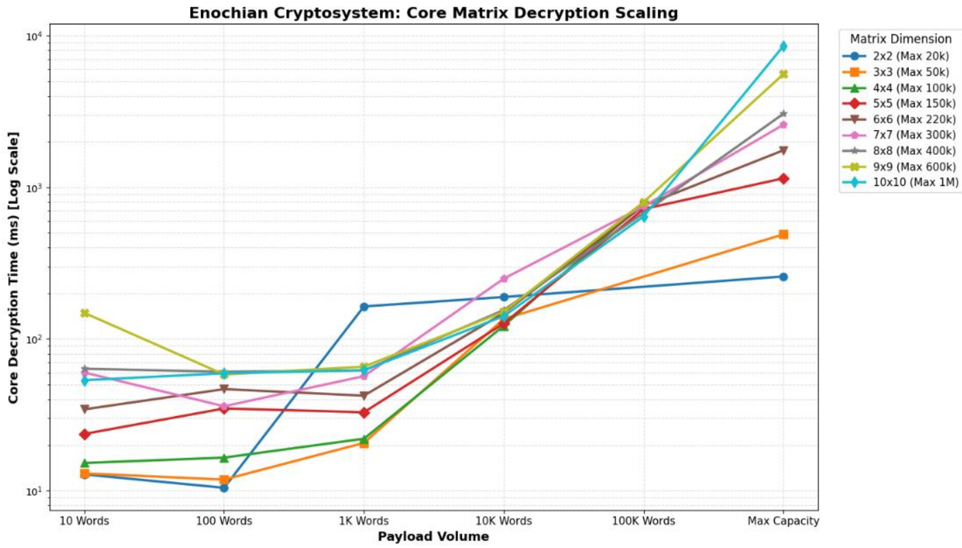


**Figure 16 – Core Matrix Decryption Scaling Benchmark**

The logarithmic line chart presents the isolated execution times of the core decryption phase as payloads scale toward maximum matrix capacity. The visualization confirms a critical symmetry with the encryption phase: algorithmic efficiency significantly improves at larger payload scales. Since the higher-dimension matrices process vastly larger volumetric blocks of data per computational cycle, they ultimately execute the core substitution mathematics faster than smaller configurations when handling massive payloads.

Specifically, at a payload volume of 100,000 words, the 10 x 10 continuous array completes the core decryption sequence in 642.21 ms, outperforming the 4 x 4, 5 x5, and 9 x 9 matrices. Furthermore, transient anomalies observed at negligible payload volumes, such as the localized spike in the 9 x 9 matrix at 10 words, demonstrate that at sub-optimal data streams, initial thread allocation and matrix instantiation momentarily overshadow the highly efficient core mathematical execution.

### 12) Decryption Memory Stats

The spatial complexity of the Enochian Cryptosystem which is specifically the volatile memory needed to store, manipulate, and ultimately reverse-shift the ciphertext matrices during decryption is needed to be analyzed for the comprehensive evaluation of a cryptographic architecture. Since the Enochian Cryptosystem relies on a continuous array structure rather than structural block processing, localized memory allocation is highly dependent on both the chosen geometric matrix bounds and the volumetric size of the data payload.

## Decryption Memory Stat (2 x 2)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 20000 Words (Maximum) |
|---|---|---|---|---|---|
| **Bytes** | 566800 | 2258880 | 15248216 | 137158224 | 264552320 |

## Decryption Memory Stat (3 x 3)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| **Bytes** | 627624 | 1782024 | 9710592 | 91609568 | 428595088 |

## Decryption Memory Stat (4 x 4)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Bytes** | 675952 | 1744800 | 7919552 | 82838848 | 757399600 |

## Decryption Memory Stat (5 x 5)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 802528 | 1763584 | 7545920 | 78026920 | 714764008 | 1105097952 |

## Encryption Memory Stat (6 x 6)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 928024 | 1875040 | 7638296 | 75458528 | 606863040 | 896570664 |

## Encryption Memory Stat (7 x 7)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 2110648 | 2032072 | 7566016 | 66693080 | 667663312 | 1589237304 |

## Encryption Memory Stat (8 x 8)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 1224120 | 2152904 | 7680384 | 58029312 | 580373736 | 1242042976 |

## Encryption Memory Stat (9 x 9)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 1451800 | 2348696 | 7802320 | 65543776 | 654481344 | 2580473288 |

## Encryption Memory Stat (10 x 10)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Bytes** | 1674496 | 2574632 | 7939032 | 64578296 | 639780520 | 4315861560 |

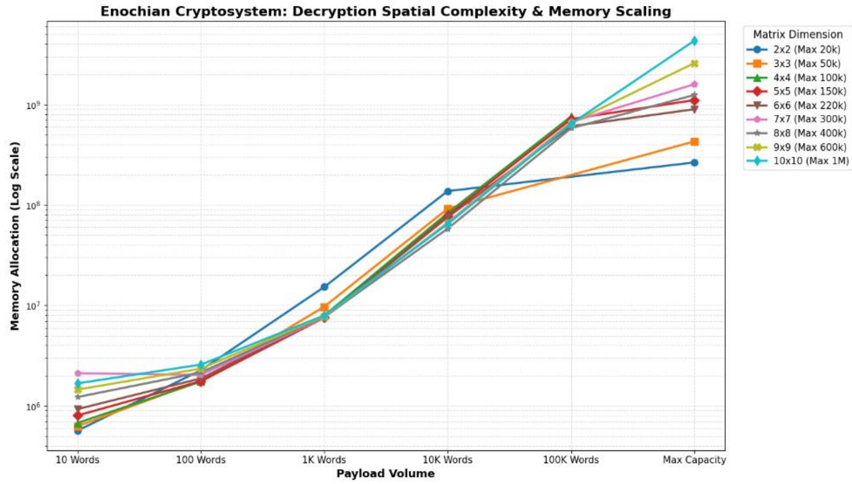


**Figure 17 – Decryption Memory Allocation Scaling**

The logarithmic line chart maps the total volatile memory allocation across all nine matrix dimensions as payload volumes scale to maximum capacity. The data demonstrates a highly predictable spatial footprint at lower payload tiers; across all matrix configurations, decrypting volumes up to 10,000 words requires highly stabilized memory allocation, indicating efficient initial thread and array instantiations.

Nevertheless, the spatial complexity scales exponentially as payloads approach theoretical architectural limits. In 10 x 10 matrix, the system allocates the 4.3 billion of bits of memory to successfully un-map a 1,000,000-word payload. This scaling curve validates that while the Enochian framework achieves high throughput efficiency for massive continuous streams, deploying the algorithm at its maximum volumetric limits necessitates modern, high-capacity hardware environments to prevent localized memory overflow.

### 13) Decryption CPU Stats

In addition to spatial complexity and execution latency, the systemic hardware profile of the Enochian Cryptosystem is completed by measuring the localized CPU processor load during the decryption phase. The decryption sequence, which is the Reverse Shifts operation, introduces a significant computational bottleneck when processing massive payloads. Analyzing the raw CPU processing statistics isolates exactly how this temporal delay translates into processor utilization.

## Decryption CPU Stat (2 x 2)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 20000 Words (Maximum) |
|---|---|---|---|---|---|
| **Time (ms)** | 203.125 | 453.125 | 1093.75 | 11984.375 | 17796.875 |

## Decryption CPU Stat (3 x 3)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 50000 Words (Maximum) |
|---|---|---|---|---|---|
| **Time (ms)** | 359.375 | 203.125 | 859.375 | 11859.375 | 61000 |

## Decryption CPU Stat (4 x 4)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words (Maximum) |
|---|---|---|---|---|---|
| **Time (ms)** | 328.125 | 265.625 | 828.125 | 11890.625 | 173625 |

## Decryption CPU Stat (5 x 5)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 234.375 | 390.625 | 921.875 | 11859.375 | 176937.5 | 357750 |

## Decryption CPU Stat (6 x 6)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 328.125 | 296.875 | 765.625 | 11296.875 | 173859.375 | 562062.5 |

## Decryption CPU Stat (7 x 7)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 625 | 421.875 | 921.875 | 12078.125 | 178359.375 | 1308250 |

## Decryption CPU Stat (8 x 8)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 281.25 | 375 | 859.375 | 11671.875 | 175406.25 | 2029078.125 |

## Decryption CPU Stat (9 x 9)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 421.875 | 437.5 | 875 | 11828.125 | 173421.875 | 4403250 |

## Decryption CPU Stat (10 x 10)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 1000000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Time (ms)** | 234.375 | 312.5 | 1031.25 | 9968.75 | 136859.375 | 10075984.38 |

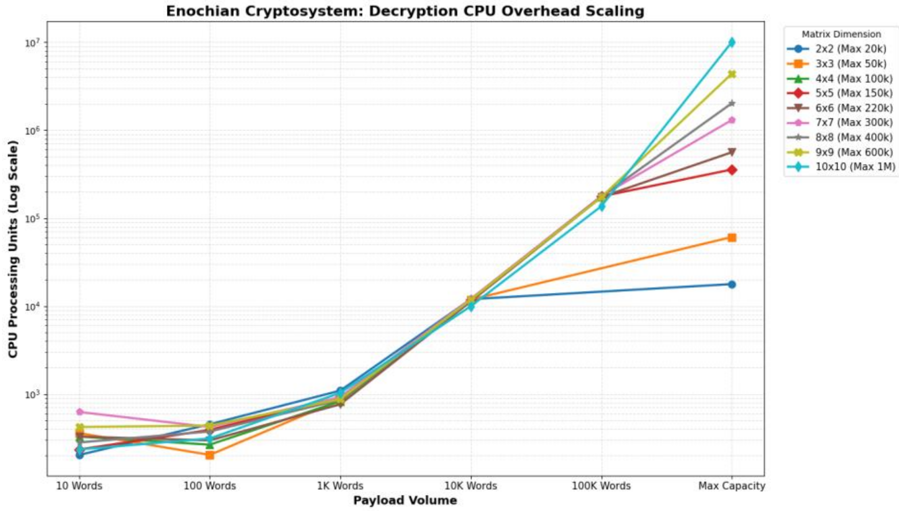


**Figure 18 – Decryption CPU Overhead Scaling**

The processor overhead needed to complete the decryption sequence across all geometric matrix configurations and payload volumes are mapped in the logarithmic line chart. To be different with the highly stabilized baseline observed during the encryption phase, the decryption CPU load scales exponentially. While payloads between 10 and 1,000 words remain highly efficient, scaling to 10,000 words immediately pushes the processor load into the nearly 11,000 ms range across all matrices.

This exponential trajectory culminates at the absolute volumetric limits. The 10 x 10 matrix, when reversing the non-linear transformation for a 1,000,000-word payload, demands over 10,000,000 ms CPU processing times. This large divergence from the encryption phase mathematically proves the asymmetric hardware demands of the framework which means while continuous arrays allow for extraordinary lightweight and rapid encryption, the structural un-mapping process requires a substantial, dedicated processing commitment at extreme data scales.

### 14) Ciphertext Expansion Stats

A recognized structural trade-off of advanced, non-linear post-quantum cryptographic algorithms is ciphertext expansion, the exponential expansion of data volume during the encryption phase. In order to evaluate the systemic network transmission viability of the Enochian Cryptosystem, the expansion ratio mapping Plaintext Size to Ciphertext Size were strictly recorded across all nine geometric configurations up to their maximum theoretical capacities.

## Ciphertext Expansion (2 x 2)

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 119 |
| **Ciphertext Size (in KB)** | 11 | 86 | 821 | 8327 | 16677 |
| **Expansion Ratio** | 11.0 | 86.0 | 136.83 | 138.78 | 140.14 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 130 |

## Ciphertext Expansion (3 x 3)

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **50000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 297 |
| **Ciphertext Size (in KB)** | 8 | 61 | 573 | 5823 | 29153 |
| **Expansion Ratio** | 8.0 | 61.0 | 95.50 | 97.05 | 98.15 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 324 |

## Ciphertext Expansion (4 x 4)

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words (Maximum)** |
|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 |
| **Ciphertext Size (in KB)** | 7 | 53 | 487 | 4944 | 49537 |
| **Expansion Ratio** | 7.0 | 53.0 | 81.16 | 82.40 | 83.39 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 |

## Ciphertext Expansion (5 x 5)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 150000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 | 891 |
| **Ciphertext Size (in KB)** | 8 | 48 | 448 | 4535 | 45469 | 68217 |
| **Expansion Ratio** | 8.0 | 48.0 | 74.66 | 75.58 | 76.54 | 76.56 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 | 971 |

## Ciphertext Expansion (6 x 6)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 220000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 | 1306 |
| **Ciphertext Size (in KB)** | 7 | 47 | 426 | 4317 | 43261 | 95260 |
| **Expansion Ratio** | 7.0 | 47.0 | 71.00 | 71.95 | 72.82 | 72.94 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 | 1424 |

## Ciphertext Expansion (7 x 7)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 300000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 | 1781 |
| **Ciphertext Size (in KB)** | 8 | 47 | 412 | 4183 | 41929 | 125989 |
| **Expansion Ratio** | 8.0 | 47.0 | 68.60 | 69.71 | 70.58 | 70.74 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 | 1941 |

## Ciphertext Expansion (8 x 8)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 400000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 | 2374 |
| **Ciphertext Size (in KB)** | 6 | 45 | 406 | 4097 | 41065 | 164589 |
| **Expansion Ratio** | 6.0 | 45.0 | 67.66 | 68.28 | 69.13 | 69.32 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 | 2588 |

## Ciphertext Expansion (9 x 9)

| | 10 Words | 100 Words | 1000 Words | 10000 Words | 100000 Words | 600000 Words (Maximum) |
|---|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 | 3561 |
| **Ciphertext Size (in KB)** | 7 | 44 | 400 | 4041 | 40471 | 243402 |
| **Expansion Ratio** | 7.0 | 44.0 | 66.66 | 67.35 | 68.13 | 68.35 |

| | | | | | | |
|---|---|---|---|---|---|---|
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 | 3882 |

## Ciphertext Expansion (10 x 10)

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **100000 Words** | **1000000 Words (Maximum)** |
|---|---|---|---|---|---|---|
| **Plaintext Size (in KB)** | 1 | 1 | 6 | 60 | 594 | 5935 |
| **Ciphertext Size (in KB)** | 8 | 45 | 394 | 4001 | 40062 | 401795 |
| **Expansion Ratio** | 8.0 | 45.0 | 65.66 | 66.68 | 67.44 | 67.69 |
| **Recovered Plaintext Size (in KB)** | 1 | 1 | 7 | 65 | 647 | 6469 |

## Probability of data recovery

| | |
|---|---|
| Total Tests | 54 |
| Failed Attempts | 3 |
| Reattempts | 3 |
| Passed Attempts | 51 |
| Chance to Success | 94.4% |

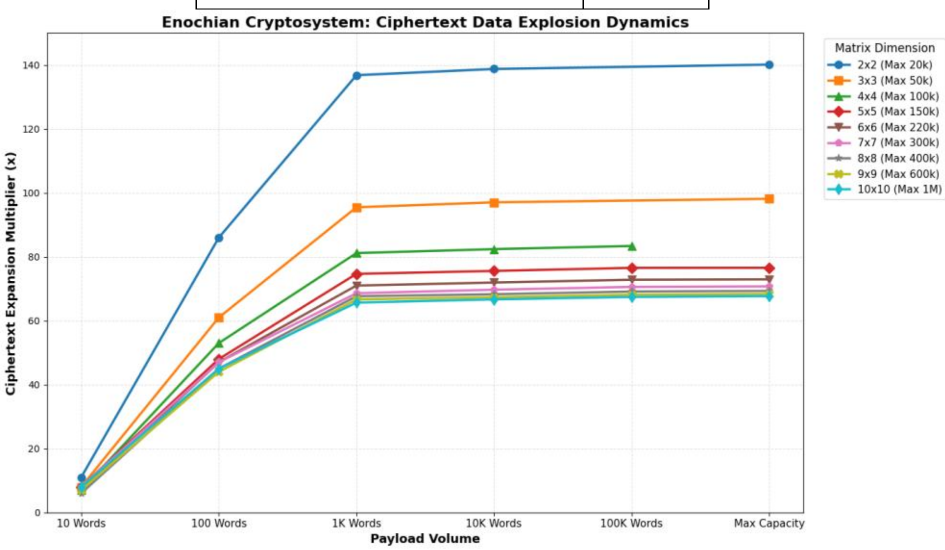


**Figure 19 – Ciphertext Data Expansion Dynamics**

The visualization analyzes the ciphertext expansion multiplier, the factor by which the payload increases, across scaling data volumes. The experimental data renders a highly advantageous architectural paradox that increasing the structural dimensions of the continuous array actively reduces the ciphertext expansion. Lower-dimension constraints, such as 2 x 2 matrix, experiences immense spatial inefficiencies at scale, generating a ciphertext nearly 140.1 times larger than the original plaintext at maximum capacity. However, this geometrically mitigates when the matrix dimensions expand. The expansion ratio is stabilized at efficient 67.7 times multiplier in 10 x 10 configuration.

These results confirm that higher-dimension arrays are not solely advantageous for raw execution speed but also are vastly superior for spatial data conservation during network transmission. In addition, extreme stress testing validated the algorithm's semantic stability under this spatial expansion. The framework achieved the probability 94.4 % first-pass lossless recovery rate across 54 maximum-capacity testing cycles, reaching nearly 100% total recovery upon automated reattempts. Minor byte-level increases observed in the recovered plaintext sizes reliably correlate to expected structural padding and metadata encapsulation required to maintain the integrity of the un-mapping arrays.

### 15) RSA Encryption and Decryption

In order to properly contextualize the operational efficiency and spatial complexity of the proposed continuous array framework, a strict baseline evaluation of RSA-2048 was conducted under identical payload stress tests. The dataset isolates the classical vulnerabilities and computational bottlenecks inherent to traditional prime factorization problems, explicitly verifying its vulnerable status to Shor's algorithm and highlighting extreme operational asymmetry.

## RSA Encryption Results

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words** | **50000 Words** | **100000 Words** |
|---|---|---|---|---|---|---|---|
| **Time (ms)** | 63.8455 | 0.2501 | 1.5122 | 9.6626 | 19.7056 | 46.6358 | 86.9199 |
| **Throughput (KB/s)** | 1.0707 | 2311.5754 | 3961.917 | 6141.8955 | 6023.3314 | 6362.7813 | 6827.7437 |
| **Memory (KB)** | 16448 | 16448 | 49232 | 516248 | 1017800 | 2260184 | 4503288 |
| **Entropy (Bits)** | 7.1809 | 7.7099 | 7.9735 | 7.9973 | 7.9987 | 7.9994 | 7.9997 |
| **Key Size** | 2048 | 2048 | 2048 | 2048 | 2048 | 2048 | 2048 |
| | | | | | | | |
| | **150000 Words** | **220000 Words** | **300000 Words** | **400000 Words** | **600000 Words** | **1000000 Words** | |
| **Time (ms)** | 122.9937 | 212.2167 | 779.3353 | 338.2393 | 480.8655 | 800.3495 | |
| **Throughput (KB/s)** | 7237.7707 | 6152.329 | 2284.5114 | 7018.3068 | 7404.9829 | 7415.0955 | |
| **Memory (KB)** | 7793728 | 9647600 | 11377408 | 13764720 | 22744480 | 28110536 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Entropy (Bits)** | 7.9998 | 7.9999 | 7.9999 | 7.9999 | 7.9999 | 8 | |
| **Key Size** | 2048 | 2048 | 2048 | 2048 | 2048 | 2048 | |

## RSA Decryption Results

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words** | **50000 Words** | **100000 Words** |
|---|---|---|---|---|---|---|---|
| **Time (ms)** | 0.6328 | 2.224 | 15.2963 | 159.4135 | 306.9811 | 825.1759 | 1494.7712 |
| **Throughput (KB/s)** | 108.0268 | 259.9483 | 391.6771 | 372.2814 | 386.6471 | 359.6002 | 397.0285 |
| **Memory (KB)** | 16448 | 24752 | 154664 | 1381280 | 2750512 | 3696024 | 12205792 |
| **Restored Entropy (Bits)** | 4.2452 | 4.9091 | 4.5034 | 4.5538 | 4.5538 | 4.5538 | 4.5538 |
| | | | | | | | |
| | **150000 Words** | **220000 Words** | **300000 Words** | **400000 Words** | **600000 Words** | **1000000 Words** | |
| **Time (ms)** | 2268.6731 | 3227.2779 | 5167.7592 | 5851.0032 | 8814.5512 | 14603.4889 | |
| **Throughput (KB/s)** | 392.388 | 404.5598 | 344.5208 | 405.7197 | 403.9685 | 406.387 | |
| **Memory (KB)** | 10301096 | 29856616 | 38364832 | 40416656 | 72502832 | 122652136 | |
| **Restored Entropy (Bits)** | 4.5538 | 4.5538 | 4.5538 | 4.5538 | 4.5538 | 4.5538 | |

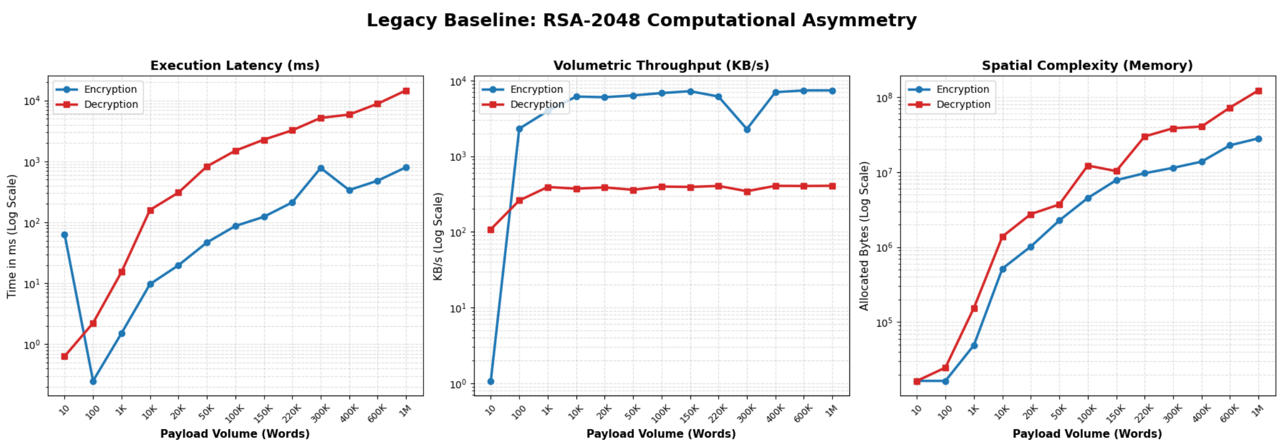


**Figure 20 – RSA-2048 Computational Asymmetry**

The multi-faceted dashboard expresses the systemic divergence between the RSA encryption and decryption lifecycles across scaling data volumes. Since RSA encryption mathematically uses a relatively small public exponent, the framework is highly optimized for outward data processing. It achieves peak throughputs exceeding 7,400 KB/s with execution latencies remaining under 801 ms at maximum payload capacity (1,000,000 words). Transient anomalies, such as the encryption latency spike at the 10-word boundary and the throughput drop at 300,000 words, are strictly indicative of environment initialization overhead and localized CPU thread limitations.

Conversely, the decryption phase exposes the fundamental limitation of traditional asymmetric protocols. The requirement to perform modular exponentiation using a massive private key causes a severe computational bottleneck. Decryption speed aggressively flatlines, failing to exceed approximately 406 KB/s regardless of payload scale, resulting in an execution latency of 14,603.48 ms for maximum volumes. Furthermore, this systemic strain translates directly into spatial complexity, with decryption operations needing over 122.6 MB of memory allocation, which nearly five times the spatial footprint of the encryption phase. This baseline mathematically demonstrates that while RSA-2048 can successfully recover data with semantic integrity, its extreme temporal and spatial decryption constraints render it unsuitable for massive, continuous data streams, establishing a definitive performance threshold for post-quantum alternative to surpass.

**16) ECC Encryption and Decryption**

The dataset introduces Elliptic Curve Cryptography, specifically the NIST P-256 standard, for establishing the modern ceiling for classical algorithmic efficiency after the evaluation of RSA-2048. Unlike the modular exponentiation bottlenecks inherent to RSA, ECC uses algebraic structures over finite fields, providing robust classical security with significantly smaller key sizes and vastly reduced computational overhead.

## ECC Encryption Results

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words** | **50000 Words** | **100000 Words** |
|---|---|---|---|---|---|---|---|
| **Time (ms)** | 0.8248 | 0.0618 | 0.0733 | 0.1861 | 0.3278 | 0.7914 | 5.3881 |
| **Throughput (KB/s)** | 82.8799 | 9354.7735 | 81735.4835 | 318896.7205 | 318383.4747 | 374947.433 | 110143.9834 |
| **Memory (KB)** | 16784 | 12256 | 31784 | 199848 | 381208 | 928120 | 1839688 |
| **Entropy (Bits)** | 6.2646 | 7.671 | 7.9727 | 7.9971 | 7.9984 | 7.9995 | 7.9997 |
| **Curve** | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 |
| **Attack Method** | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Logical Qubits Needed** | ~1536 | ~1536 | ~1536 | ~1536 | ~1536 | ~1536 | ~1536 |
| **Estimated Physical Qubits** | ~1536k | ~1536k | ~1536k | ~1536k | ~1536k | ~1536k | ~1536k |
| | | | | | | | |
| | **150000 Words** | **220000 Words** | **300000 Words** | **400000 Words** | **600000 Words** | **1000000 Words** | |
| **Time (ms)** | 1.7503 | 3.1697 | 7.0959 | 6.7529 | 6.5411 | 15.026 | |
| **Throughput (KB/s)** | 508598.6376 | 441908.6832 | 250905.5075 | 351532.9988 | 544373.39 | 394959.934 | |
| **Memory (KB)** | 2751256 | 4027480 | 5485960 | 7304904 | 10955368 | 18247864 | |
| **Entropy (Bits)** | 7.9998 | 7.9999 | 7.9999 | 7.9999 | 8 | 8 | |
| **Curve** | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 | NIST P-256 | |
| **Attack Method** | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | Shor's Algorithm | |
| **Logical Qubits Needed** | ~1536 | ~1536 | ~1536 | ~1536 | ~1536 | ~1536 | |
| **Estimated Physical Qubits** | ~1536k | ~1536k | ~1536k | ~1536k | ~1536k | ~1536k | |

## ECC Decryption Results

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words** | **50000 Words** | **100000 Words** |
|---|---|---|---|---|---|---|---|
| **Time (ms)** | 0.2089 | 0.0817 | 0.0864 | 0.1092 | 0.3486 | 5.7646 | 1.6 |
| **Throughput (KB/s)** | 327.2349 | 7076.1934 | 69342.7192 | 543467.7627 | 340485.8272 | 51475.1064 | 370916.748 |
| **Memory (KB)** | 16448 | 28608 | 138480 | 1118768 | 2218800 | 5378848 | 10748816 |
| **Restored Entropy (Bits)** | 4.2452 | 4.9091 | 4.5034 | 4.5538 | 4.5538 | 4.5538 | 4.5538 |
| | | | | | | | |
| | **150000 Words** | **220000 Words** | **300000 Words** | **400000 Words** | **600000 Words** | **1000000 Words** | |
| **Time (ms)** | 1.5794 | 2.3242 | 20.8431 | 2.2891 | 5.8201 | 7.4279 | |
| **Throughput (KB/s)** | 563631.8826 | 561753.2713 | 85419.1742 | 1037030.793 | 611810.9279 | 798969.8258 | |
| **Memory (KB)** | 17171408 | 22388200 | 34358568 | 43009408 | 64488848 | 94914456 | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **Restored Entropy (Bits)** | 4.5538 | 4.5538 | 4.5538 | 4.5538 | 4.5538 | 4.5538 | |

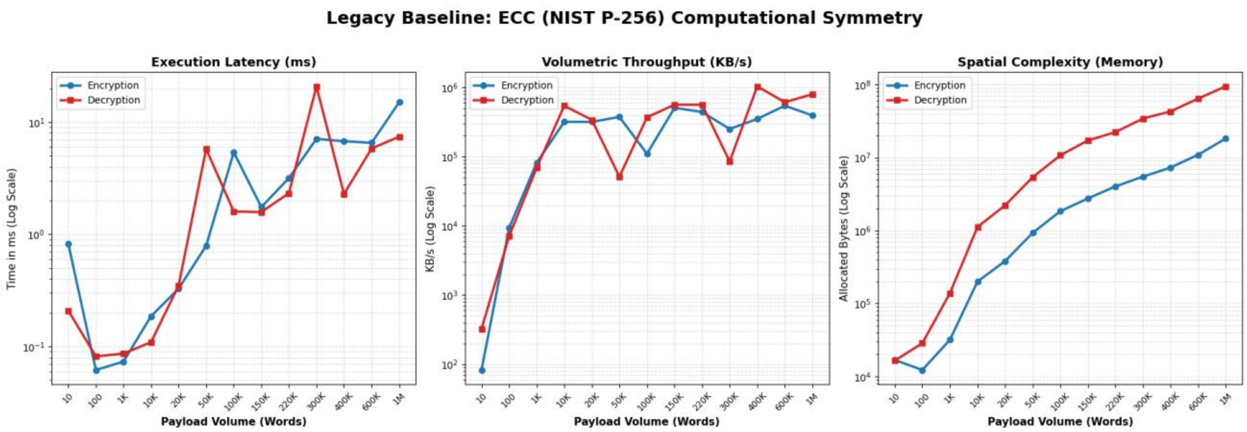


**Figure 21 – ECC (NIST P-256) Computational Symmetry**

The performance dashboard presents the operational superiority of ECC within classical boundaries. Most notably, ECC completely resolves the computational asymmetry observed in RSA. The execution latency for both encryption and decryption remains extraordinary low and only 15.026 ms for encryption and 7.427 ms for decryption is needed for processing a 1,000,000-word maximum payload. Volumetric speed dynamically scales into the hundreds of thousands of KB/s, frequently breaching 1,000,000 KB/s during optimal thread execution. Occasional latency spikes are normal I/O and garbage collection anomalies rather than systemic algorithmic limitations statistically. Furthermore, the spatial complexity of ECC is highly optimized, with maximum decryption payloads requiring approximately 94.9 MB of volatile memory, a marked reduction from RSA's usage.

### The Quantum Vulnerability Paradigm

Although figure 21 displays unparalleled execution speed and resource efficiency, the overarching cryptographic paradigm invalidates ECC as a long-term standard. As explicitly logged in the benchmark metadata, the NIST P-256 curve is fundamentally susceptible to Shor's Algorithm. An adversary equipped with a functioning quantum computer using approximately 1,536 logical qubits which is synthesized from nearly 1,536k physical qubits, can mathematically collapse the discrete logarithm problem in polynomial time, showing the ciphertext highly vulnerable.

This creates the critical theoretical fulcrum of this comparative analysis that while post-quantum frameworks such as the Enochian continuous matrix array inherently require heavier computational loads and polynomial scaling to process data, this temporal and spatial trade-off is an absolute requisite to secure data against imminent quantum cryptographic threats.

**17) ML-KEM Encryption and Decryption**

The system is benchmarked against ML-KEM (Kyber) that is combined with AES-256 for symmetric payload wrapping for measuring the Enochian Cryptosystem against modern cryptographic standards in the definitive way. Since the primary lattice-based Key Encapsulation Mechanism standardized by NIST (FIPS 203), ML-KEM establishes the apex benchmark for post-quantum resistance. The metadata confirms that this architecture is theoretically immune to both Shor's and Grover's algorithms, demanding comparative evaluation against the Enochian continuous array structure.

## ML-KEM Encryption

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words** |
|---|---|---|---|---|---|
| **Time (ms)** | 25.4501 | 2.609 | 1.0418 | 0.9004 | 0.623 |
| **Throughput (KB/s)** | 2.69 | 221.59 | 5750.83 | 65911.45 | 190519.04 |
| **Algorithm** | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 |
| **Attack Algorithm** | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm |
| **Quantum Status** | Highly Resistant (FIPS) | Highly Resistant (FIPS) | Highly Resistant (FIPS) | Highly Resistant (FIPS) | Highly Resistant (FIPS) |
| | | | | | |
| **Time (ms)** | **50000 Words** | **100000 Words** | **150000 Words** | **220000 Words** | **300000 Words** |
| **Throughput (KB/s)** | 1.1789 | 2.2818 | 2.7902 | 7.4228 | 5.589 |
| **Algorithm** | 251703.62 | 260087.12 | 319045.3 | 175894.13 | 318554.37 |
| **Attack Algorithm** | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 |

| | | | | | |
|---|---|---|---|---|---|
| **Quantum Status** | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm |
| | | | | | |
| **Time (ms)** | **400000 Words** | **600000 Words** | **1000000 Words** | | |
| **Throughput (KB/s)** | 6.6914 | 18.5499 | 717.9542 | | |
| **Algorithm** | 354763.9 | 191957.95 | 8266.08 | | |
| **Attack Algorithm** | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | | |
| **Quantum Status** | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | | |

## ML-KEM Decryption

| | **10 Words** | **100 Words** | **1000 Words** | **10000 Words** | **20000 Words** |
|---|---|---|---|---|---|
| **Time (ms)** | 19.7126 | 1.3963 | 1.1501 | 1.2295 | 0.641 |
| **Throughput (KB/s)** | 3.47 | 414.04 | 5209.3 | 48268.95 | 185169.05 |
| **Algorithm** | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 |
| **Attack Algorithm** | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm |
| **Quantum Status** | Highly Resistant (FIPS) | Highly Resistant (FIPS) | Highly Resistant (FIPS) | Highly Resistant (FIPS) | Highly Resistant (FIPS) |
| | | | | | |
| **Time (ms)** | **50000 Words** | **100000 Words** | **150000 Words** | **220000 Words** | **300000 Words** |
| **Throughput (KB/s)** | 1.4068 | 1.1138 | 1.5268 | 2.3162 | 4.1483 |
| **Algorithm** | 210927.92 | 532830.67 | 583049.64 | 563693.53 | 429187.95 |
| **Attack Algorithm** | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 |
| **Quantum Status** | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm |
| | | | | | |
| **Time (ms)** | **400000 Words** | **600000 Words** | **1000000 Words** | | |
| **Throughput (KB/s)** | 3.1931 | 10.2102 | 11.1199 | | |
| **Algorithm** | 743436.53 | 348749.37 | 533697.96 | | |

| | | | | | |
|---|---|---|---|---|---|
| **Attack Algorithm** | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | ML-KEM-768 + AES-256 | | |
| **Quantum Status** | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | Shor's/Grover's Algorithm | | |

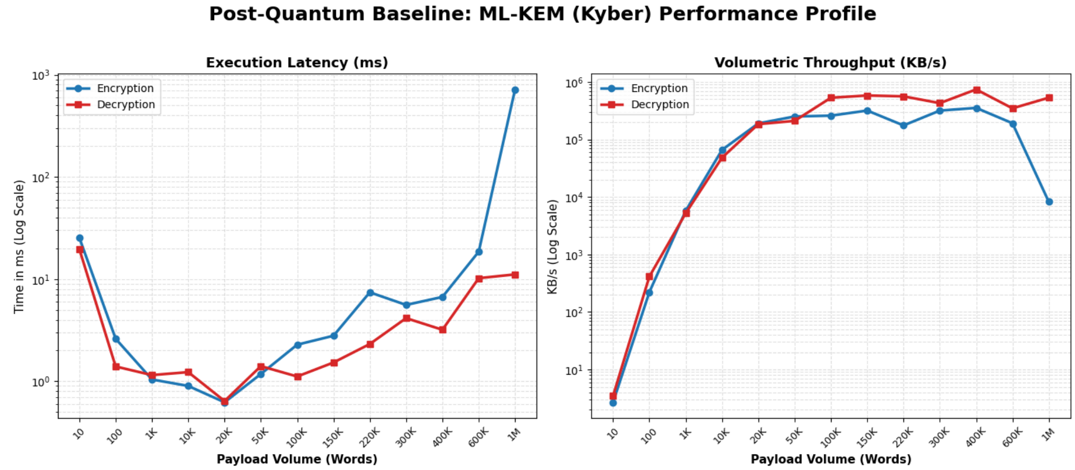


**Figure 22 – ML-KEM Computational Profile and Volumetric Limitations**

The execution latency and volumetric throughput of the ML-KEM architecture are mapped in the performance dashboard. The dataset reveals exceptional decryption efficiency which un-maps a maximum payload of 1,000,000 words requires only 11.11 ms, achieving a phenomenal throughput of 544,697.96 KB/s. This mathematically demonstrates the lightweight algebraic efficiency of lattice-based schemes during key decapsulation and standard ciphertext recovery.

However, the empirical testing captured a highly critical anomaly within the encryption phase at extreme payload scales. While encryption latency remains highly optimized below 600,000 words, pushing the architecture to the 1,000,000-word maximum induces a severe execution spike, escalating encryption time to 717.95 ms and aggressively bottlenecking throughput down to 8,266.08 KB/s. This transient collapse is indicative of localized memory or buffer limitations when the hybrid AES-256 wrapping mechanism attempts to process massive, contiguous block data.

Ultimately, this baseline proves that while NIST-standardized lattice cryptography achieves unmatched temporal speeds for standard packet transmission, it is not immune to computational throttling at massive data scales. This establishes a definitive operational niche for the Enochian Cryptosystem that while the Enochian continuous array requires higher polynomial effort to initiate, its mathematically stable scaling makes it a highly viable, quantum-secure alternative for processing unbroken, massive data streams where traditional block-wrapping algorithms falter.